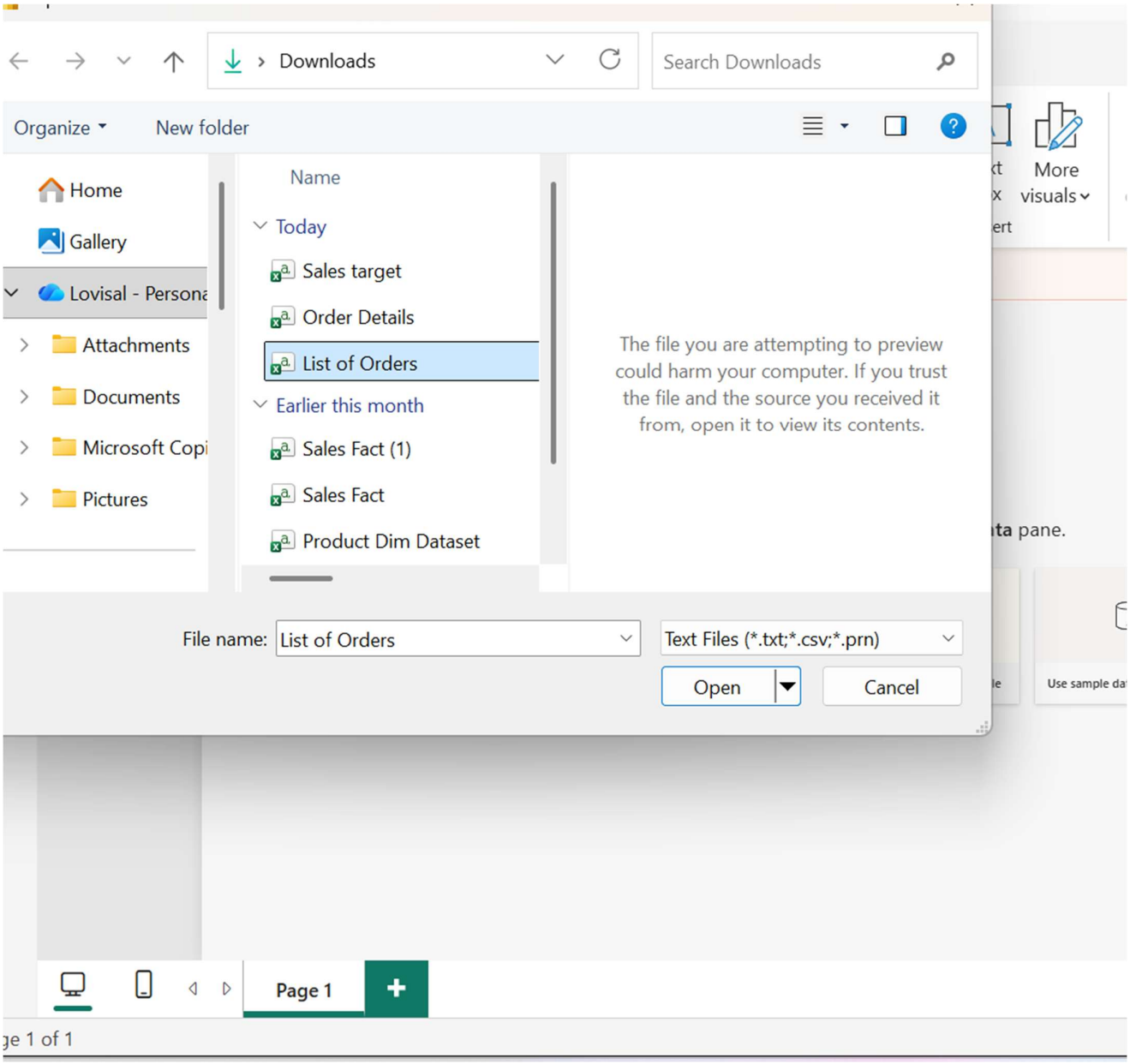
Power BI Assignment 1 – Data Transformation & Data Modeling

1. Importing the Data to power BI

In Power BI Desktop, Home > Get data > Text/CSV. In the Open window, browsed and select "List of Orders" and clicked Open to load into Power BI.



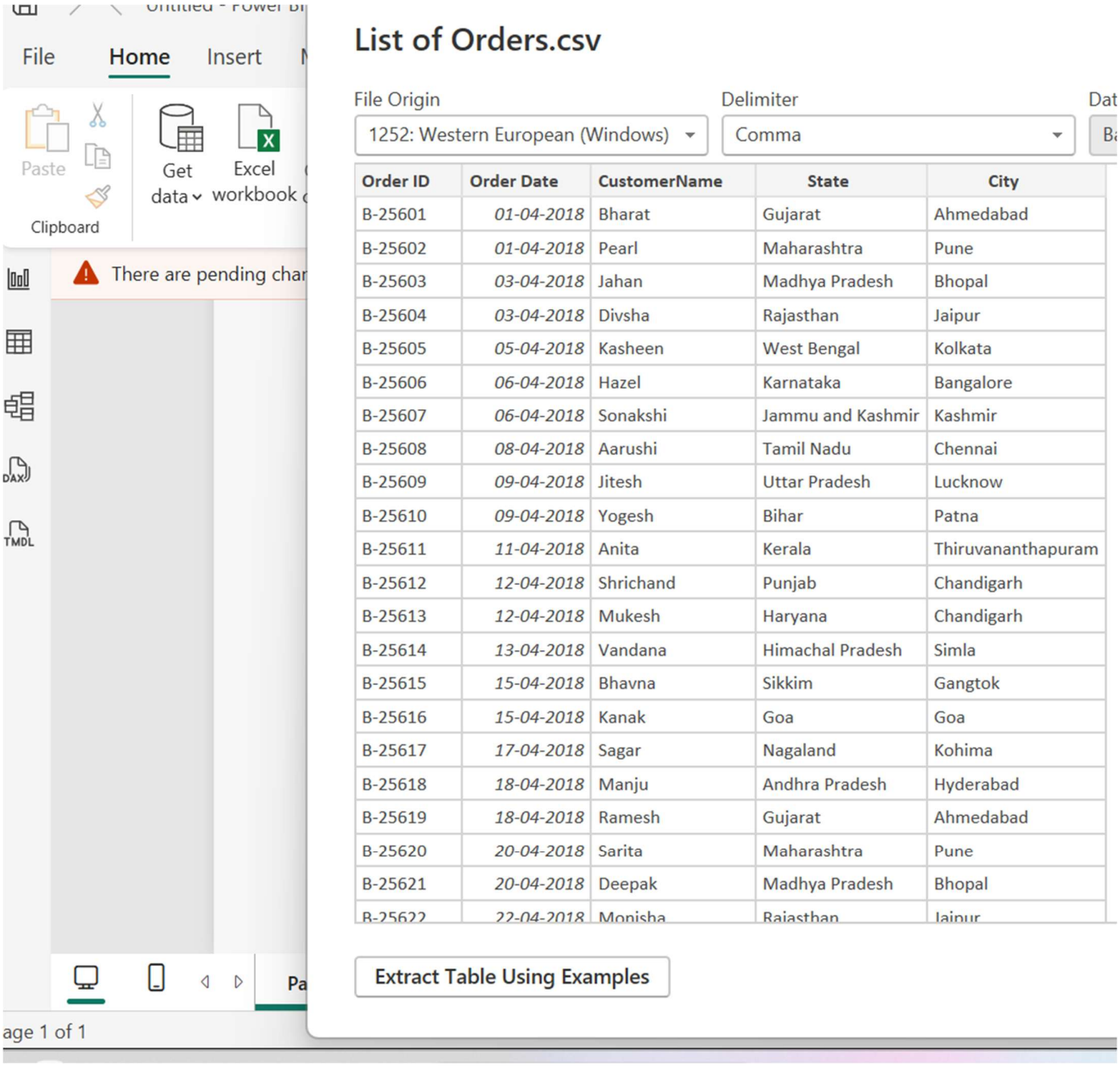
In the file preview window:

1. Verify the five columns: **Order ID**, **Order Date**, **Customer Name**, **State**, and **City**.

2. Set **Data Type Detection** to **"Based on entire dataset."**
3. Click **Transform Data** to launch the Power Query Editor.

To import the remaining files:

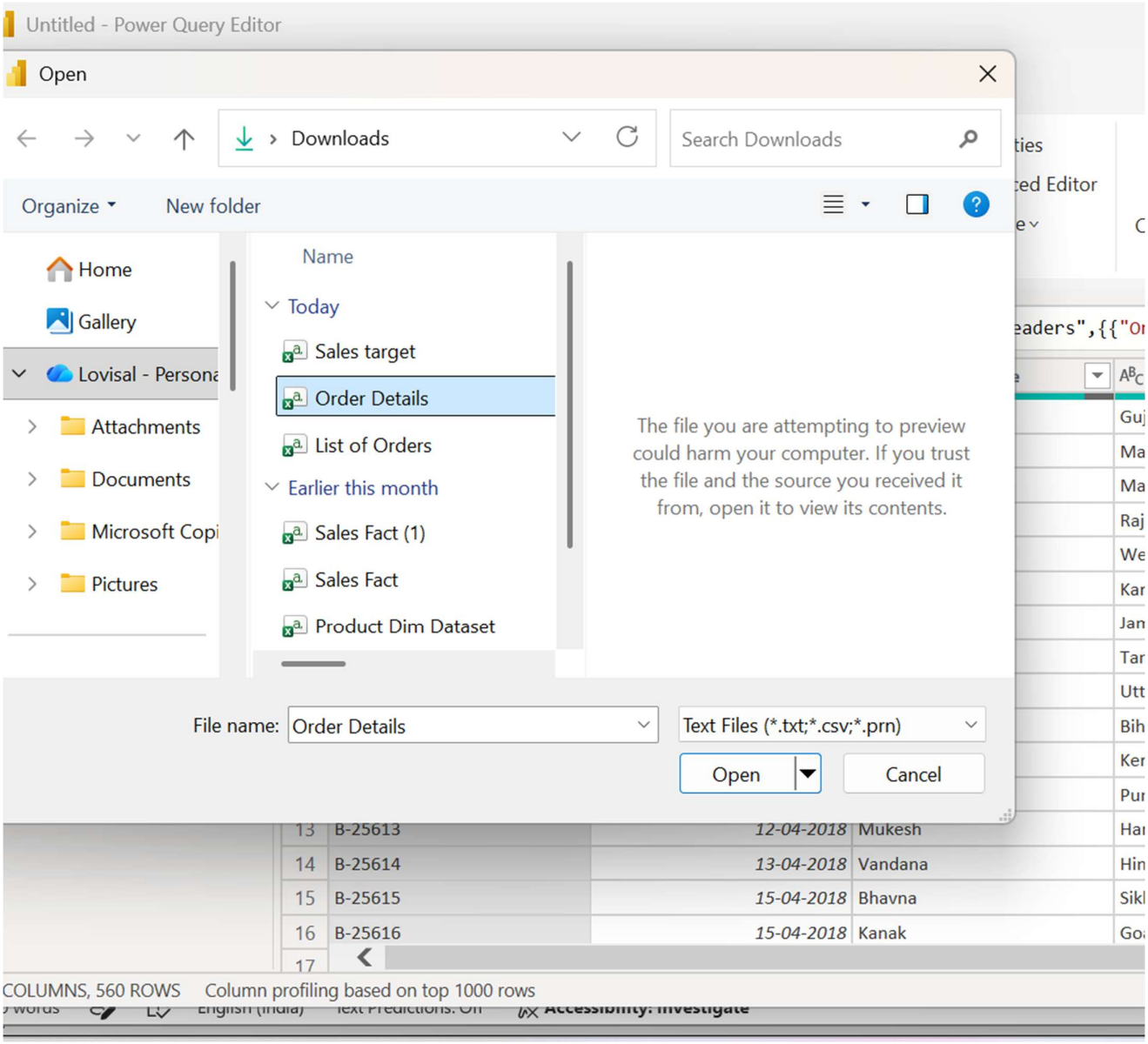
1. In the Power Query Editor, go to the **Home** tab.
2. Click **New Source** > **Text/CSV**.
3. Select and import the remaining two files into the editor.

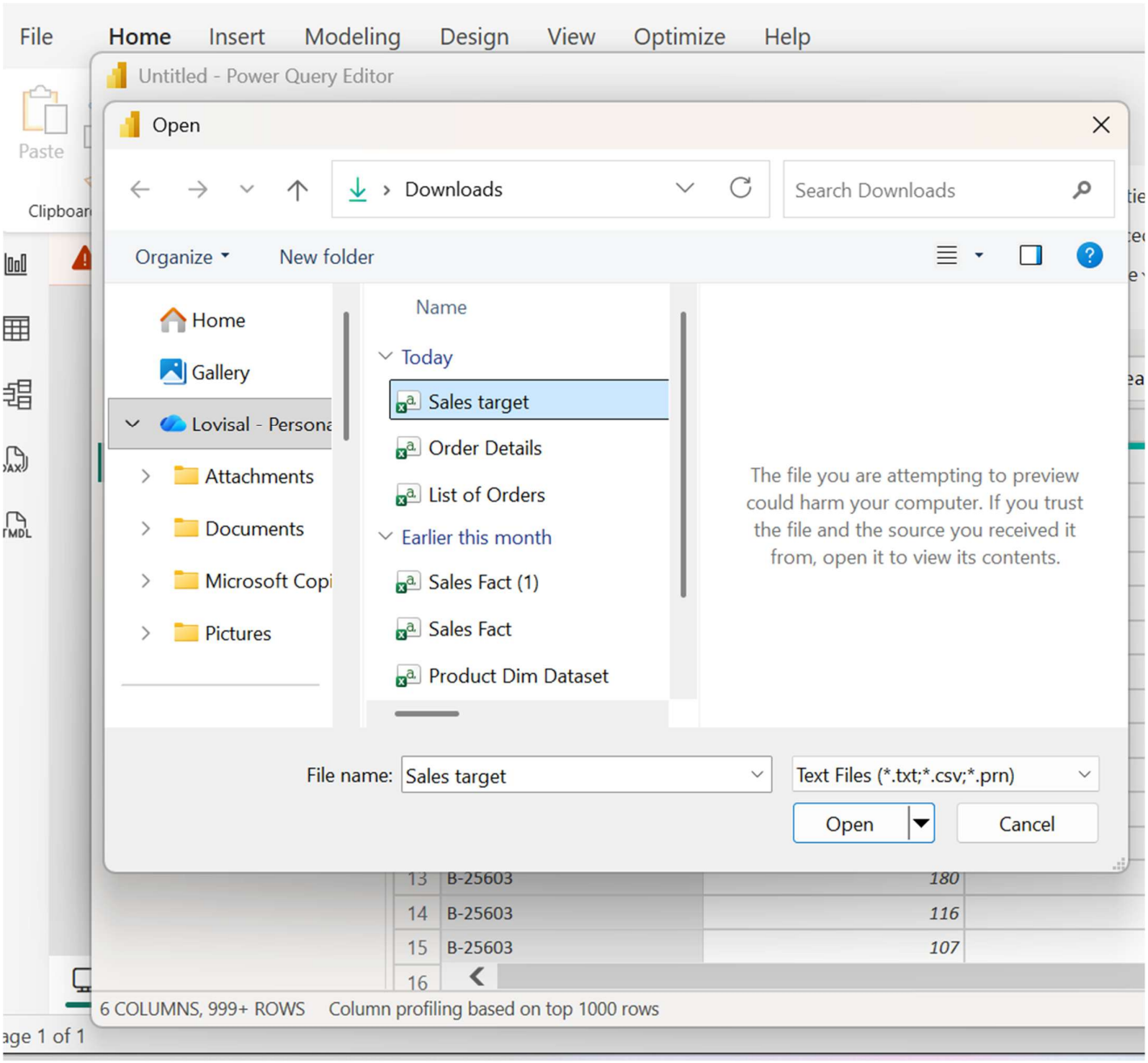


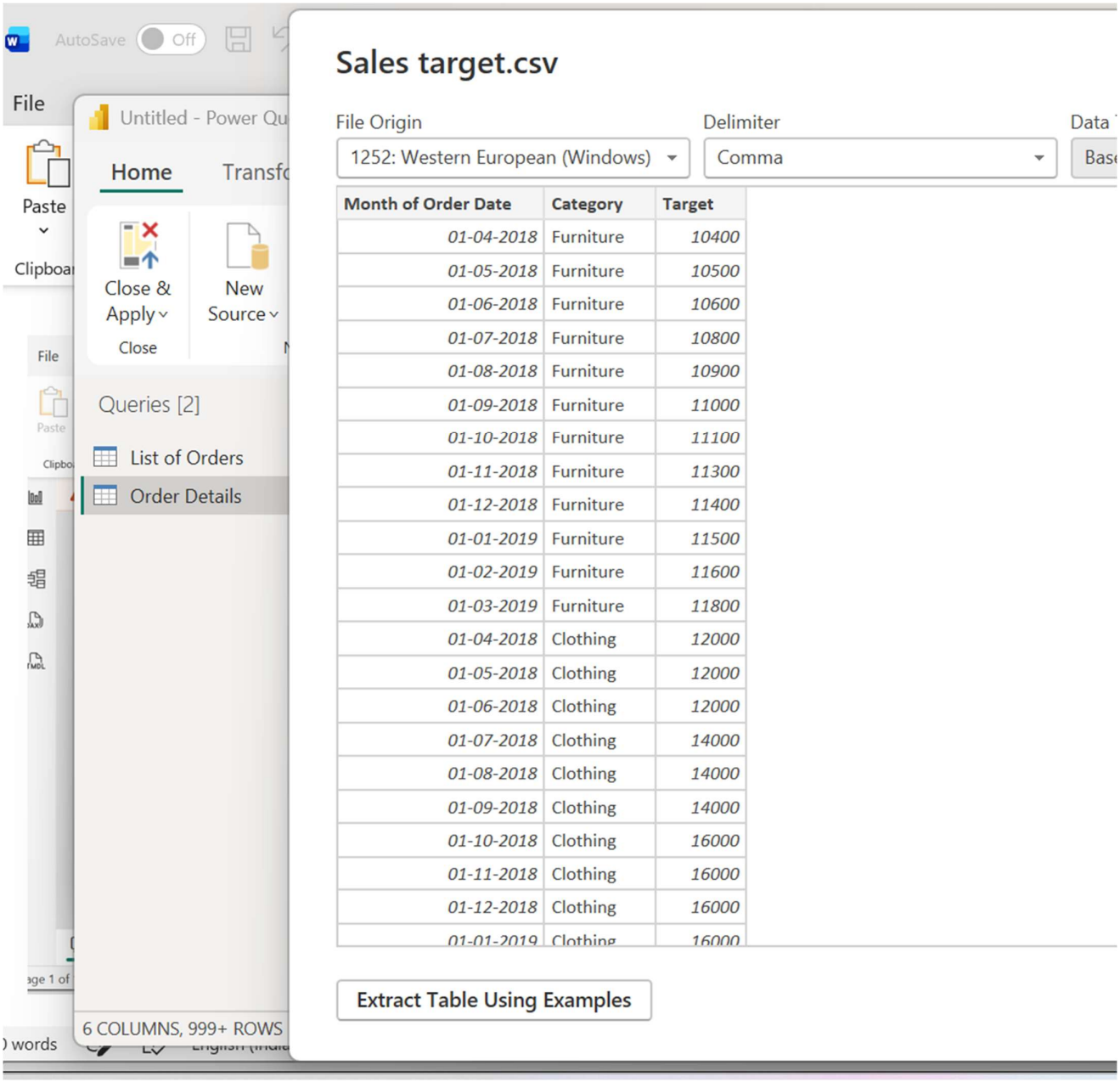
Import the Order Details and Sales Target files

1. In the **Open** window, select **Order Details** from the same folder and click **Open**.

2. Repeat this process to import the **Sales Target** file.




File
Home
Insert
Modeling
Design
View
Optimize
Help
Untitled - Power Query Editor
Paste
Clipboard
Open
Downloads
Search Downloads
Organize
New folder
Home
Gallery
Lovisal - Persona
Attachments
Documents
Microsoft Copi
Pictures
Name
Today
Sales target
Order Details
List of Orders
Earlier this month
Sales Fact (1)
Sales Fact
Product Dim Dataset
The file you are attempting to preview could harm your computer. If you trust the file and the source you received it from, open it to view its contents.
File name:
Sales target
Text Files (*.txt;*.csv;*.prn)
Open
Cancel
13 B-25603 180
14 B-25603 116
15 B-25603 107
16
6 COLUMNS, 999+ ROWS
Column profiling based on top 1000 rows
age 1 of 1

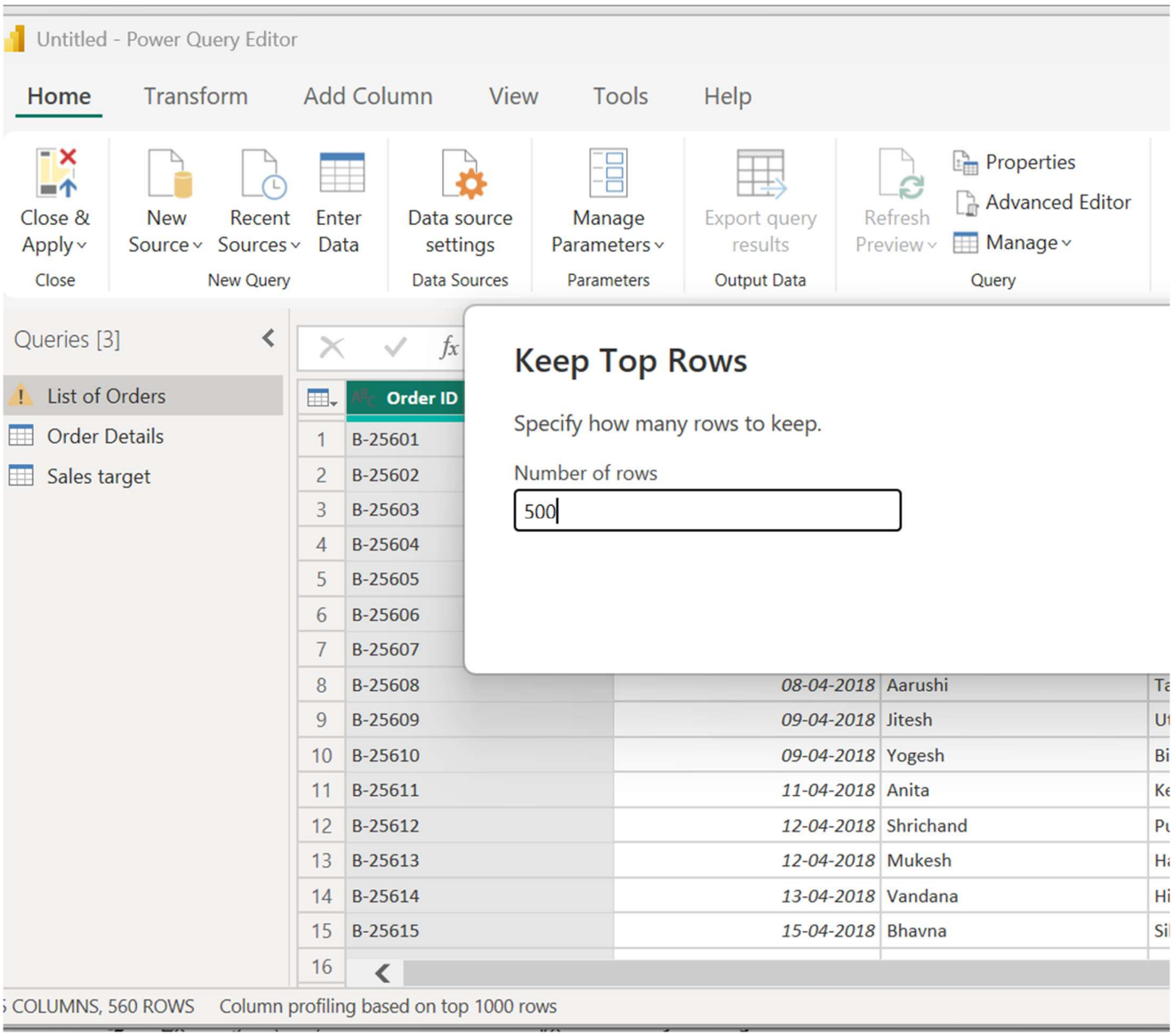


Data Transformation:

To restrict the table to 500 rows:

1. Select the **List of Orders** query from the Queries pane.
2. Go to the **Home** tab and select **Keep Rows** > **Keep Top Rows**.
3. Enter **500** as the number of rows to keep, then click **OK**.

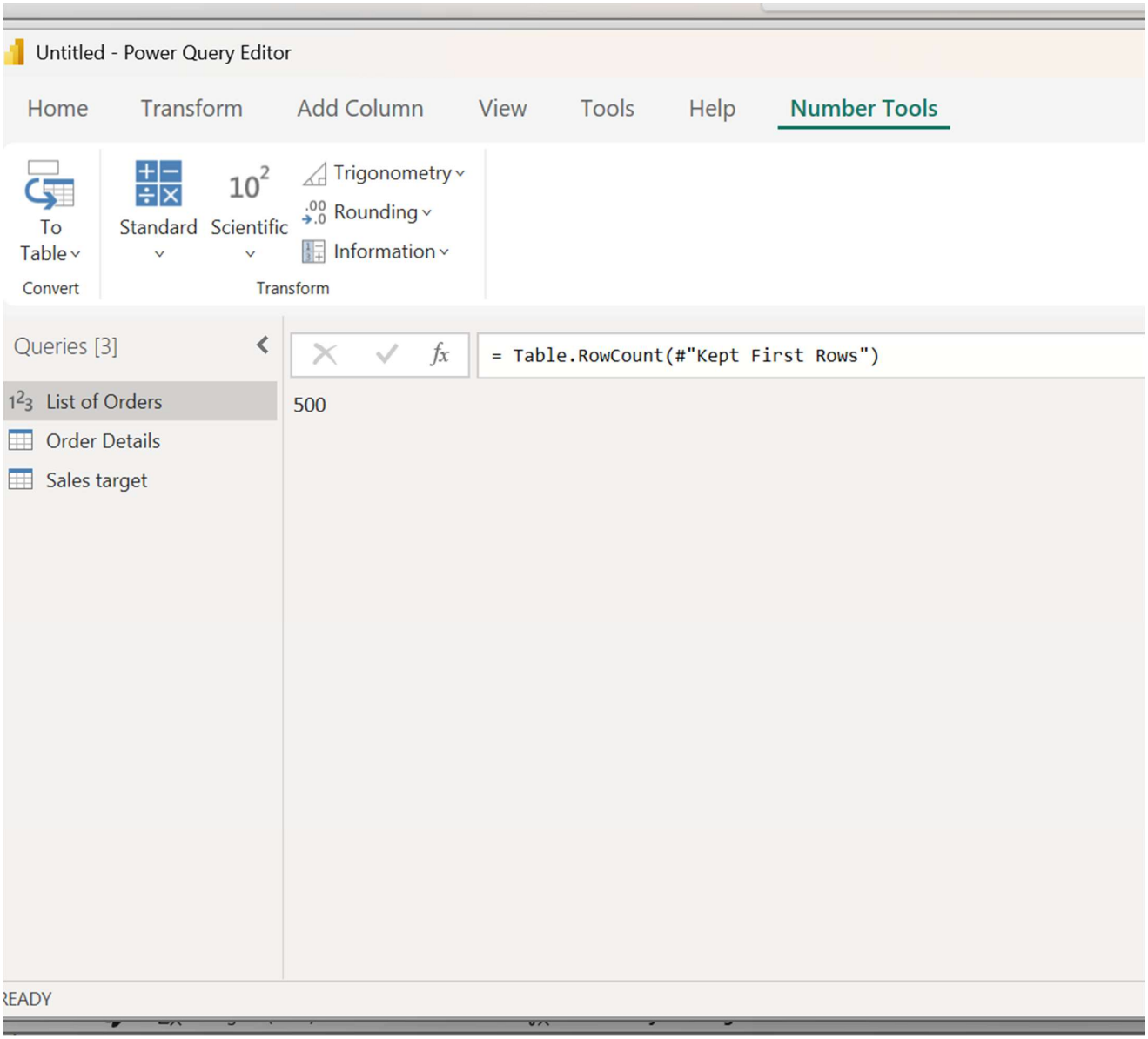
In the Keep Top Rows window, type 500 as the number of rows and click OK.



Verify the row count

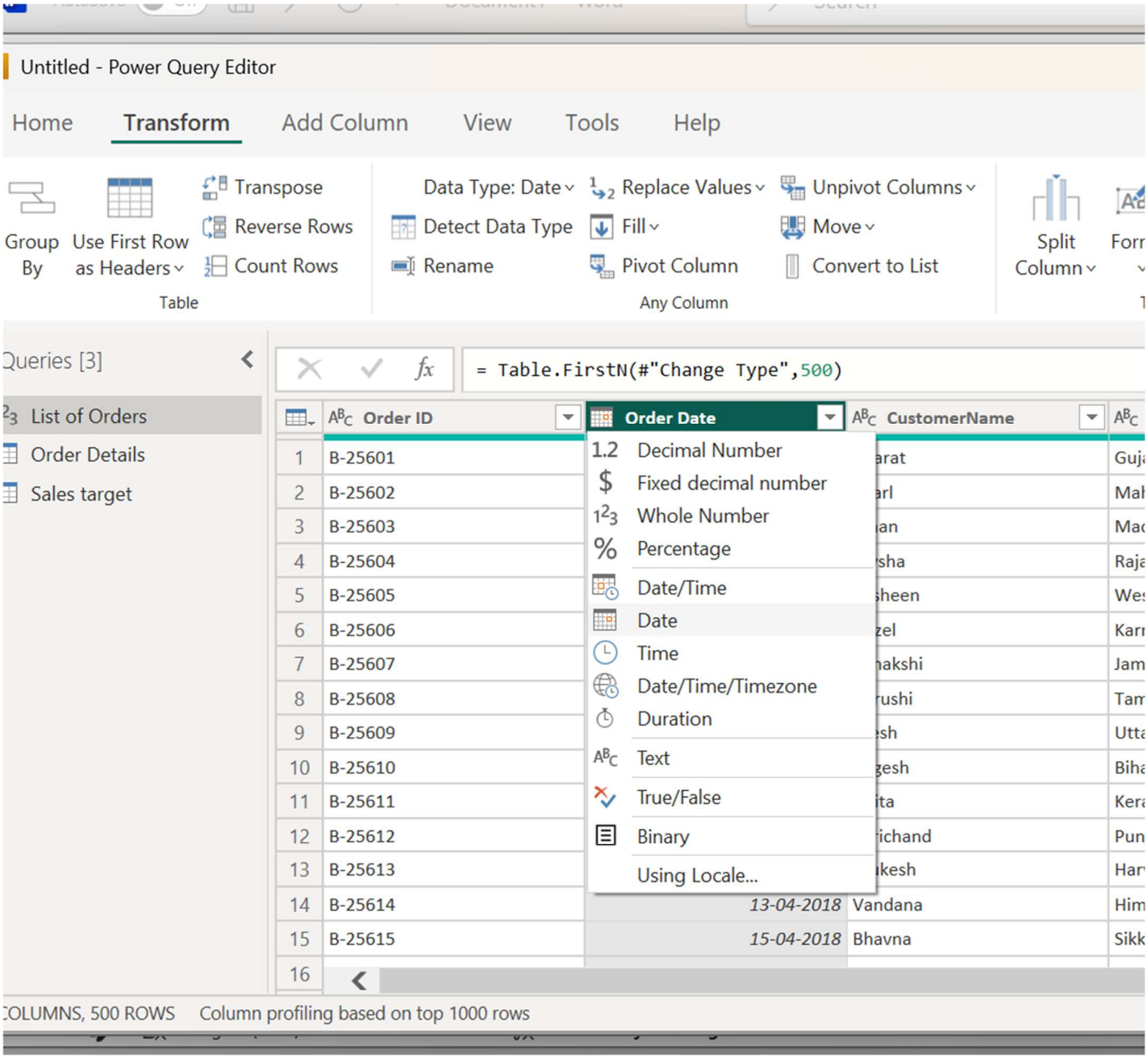
1. Go to the **Transform** tab and select **Count Rows**.
2. Confirm that the result returns **500**, verifying that only the top 500 rows remain.

3. Observe the **Applied Steps** pane, which now lists *Kept First Rows* and *Counted Rows*.
4. Since this count was only a verification step, delete the **Counted Rows** step from the **Applied Steps** pane to restore the table view.



To change the Order Date data type:

1. Click the **Data Type** icon on the left side of the **Order Date** column header.
2. Select **Date** from the drop-down menu.

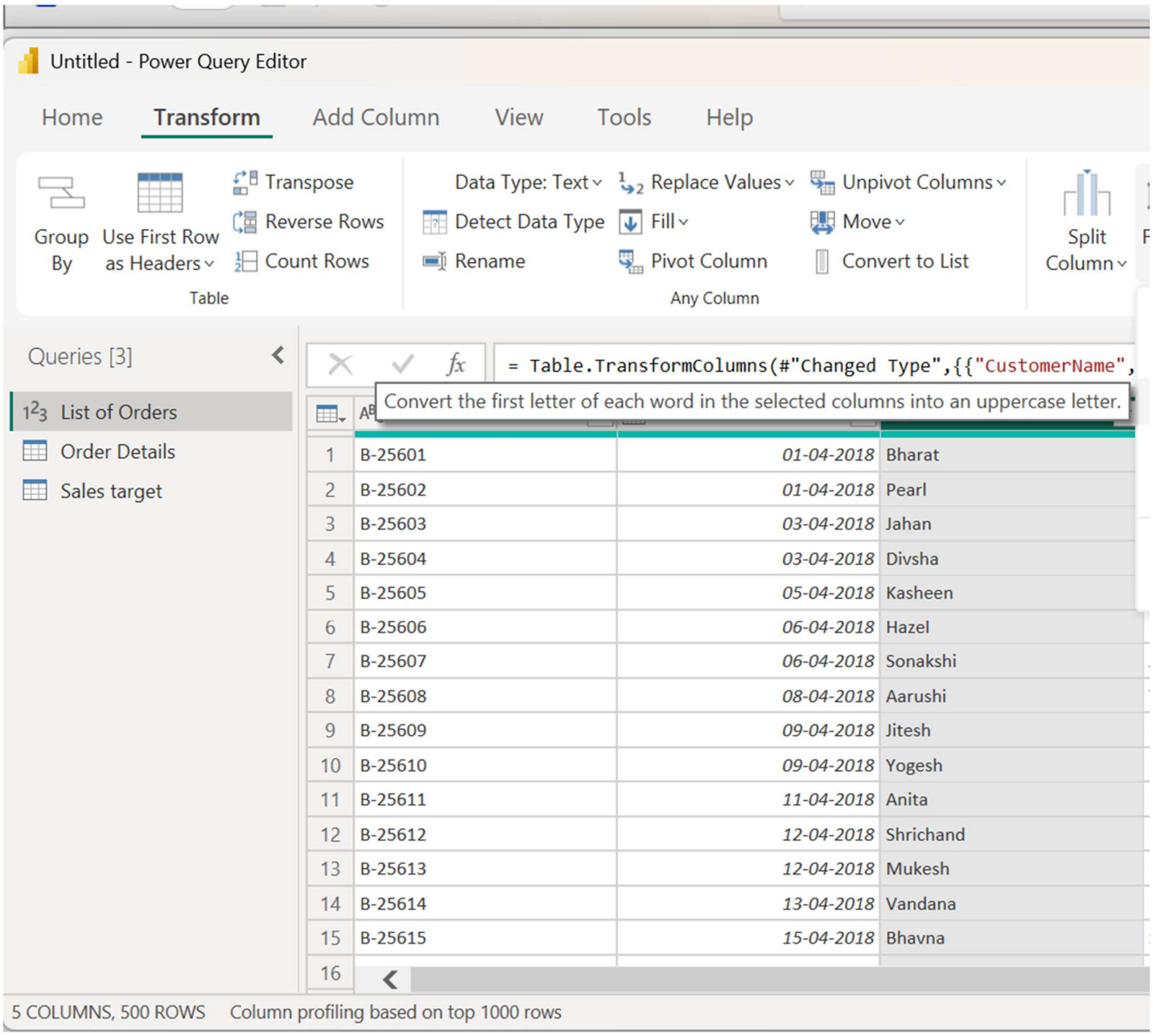


To convert Customer Name to proper case:

1. **Select the Customer Name column.**

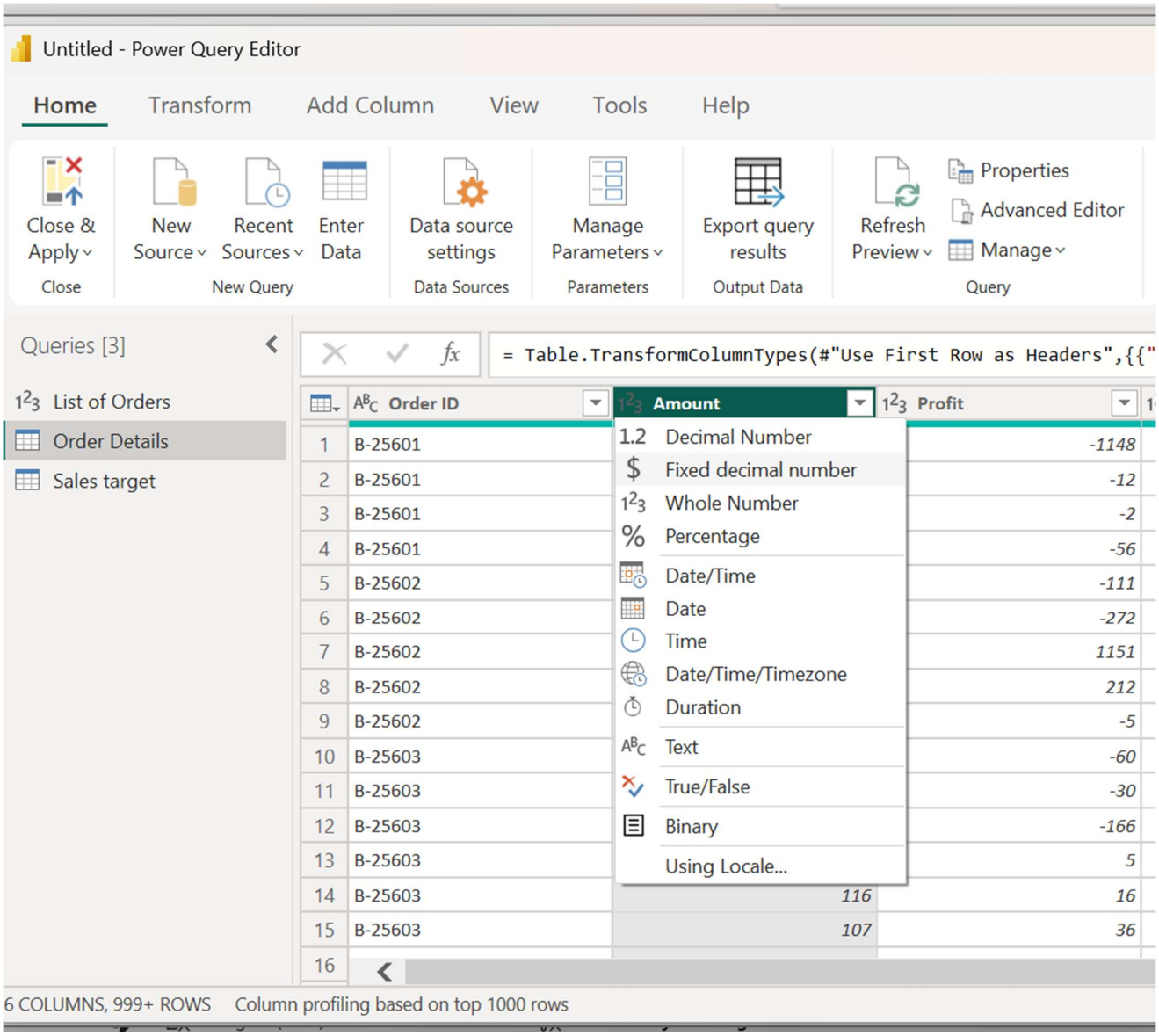
2. **Navigate to Transform > Format > Capitalize Each Word.**

This will capitalize the first letter of each word in the column.



To change the Amount column to currency/fixed decimal:

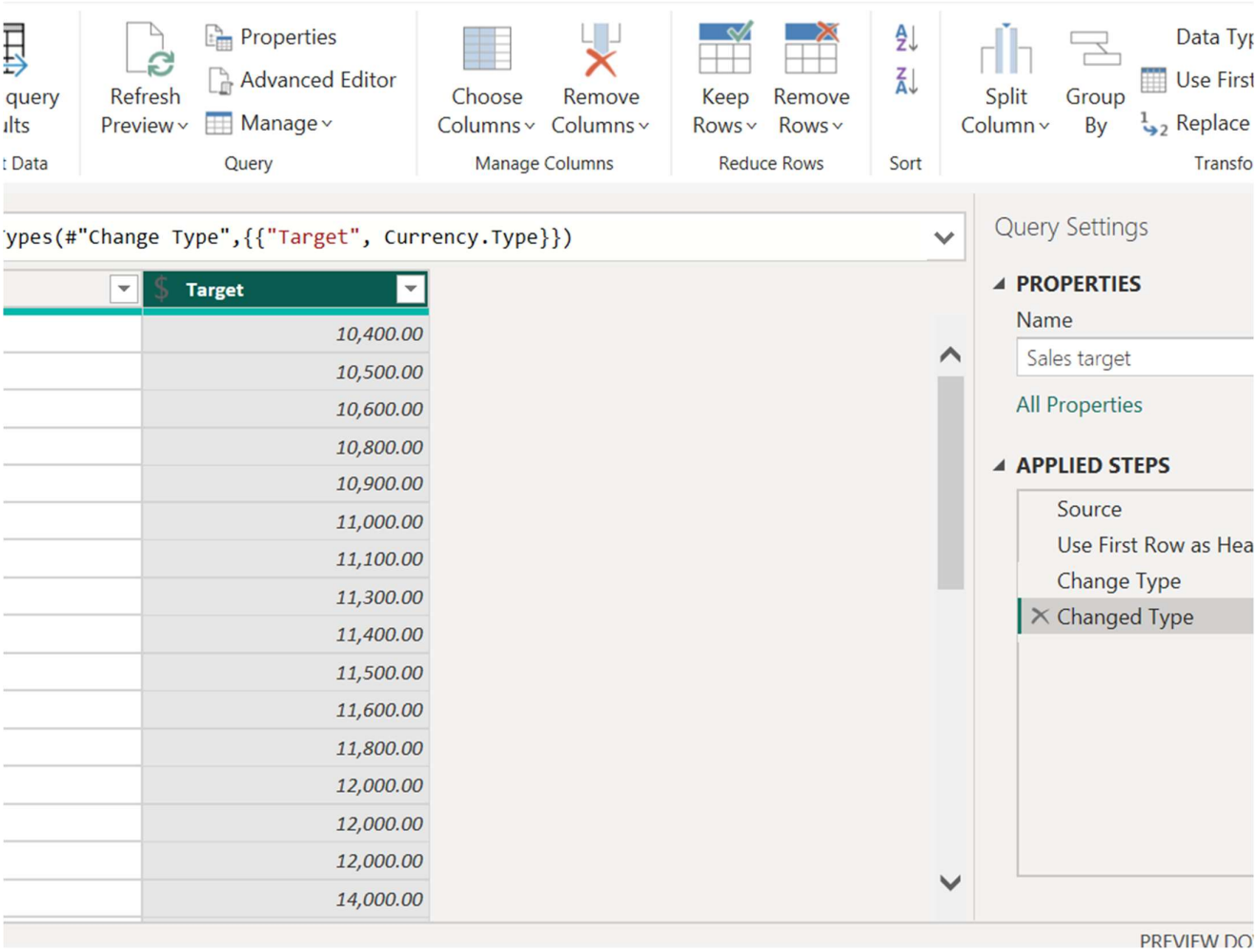
1. In the **Order Details** query, click the **Data Type** icon in the **Amount** column header.
2. Select **Fixed decimal number** from the list.



To set the Target column to Fixed Decimal Number:

1. Select the **Sales target** query.
2. Click the **Data Type** icon in the **Target** column header and choose **Fixed decimal number**.

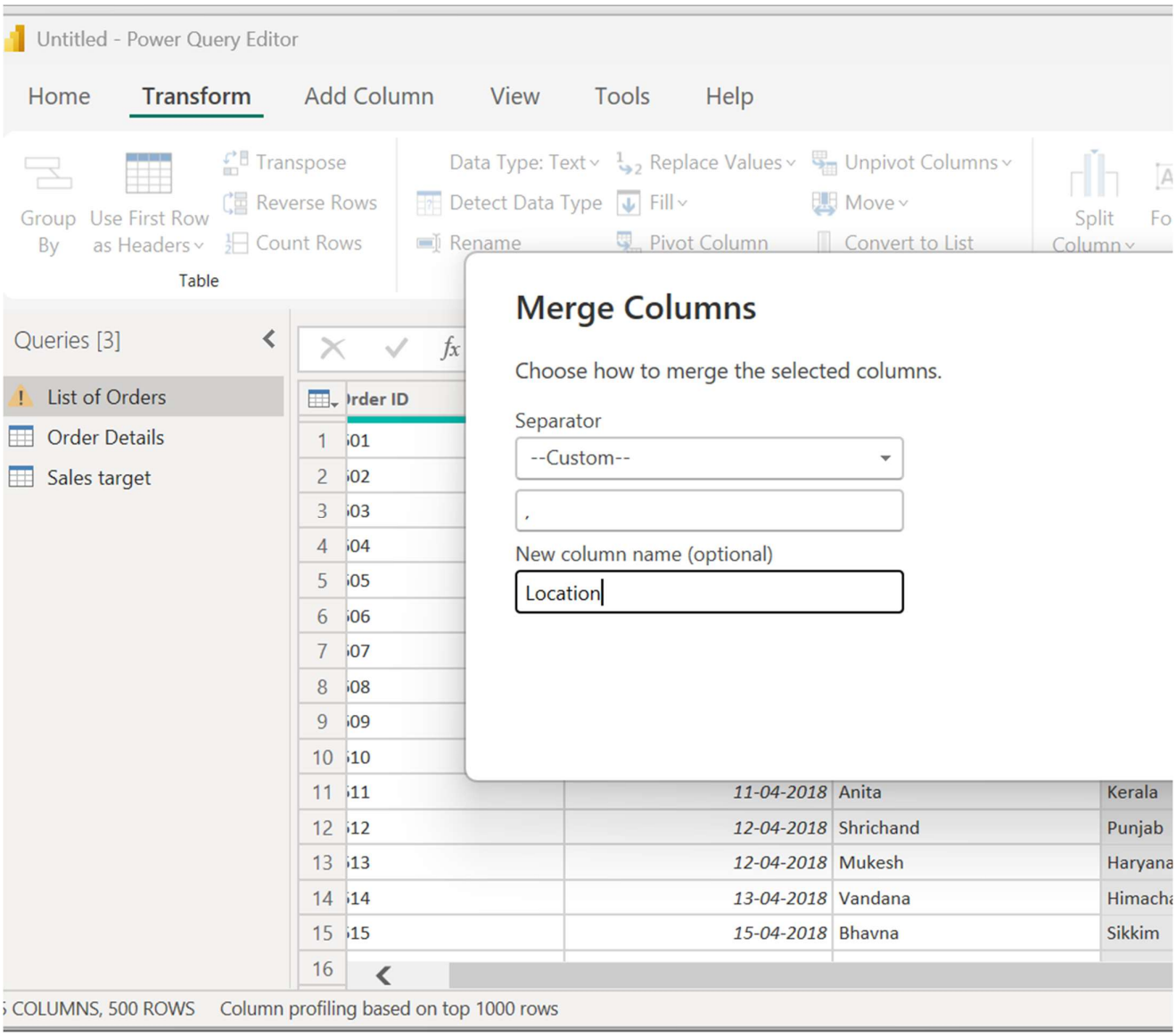
3. Verify the change by checking for the **$** icon in the header and formatted currency values (e.g., **10,400.00**).



To merge State and City into Location:

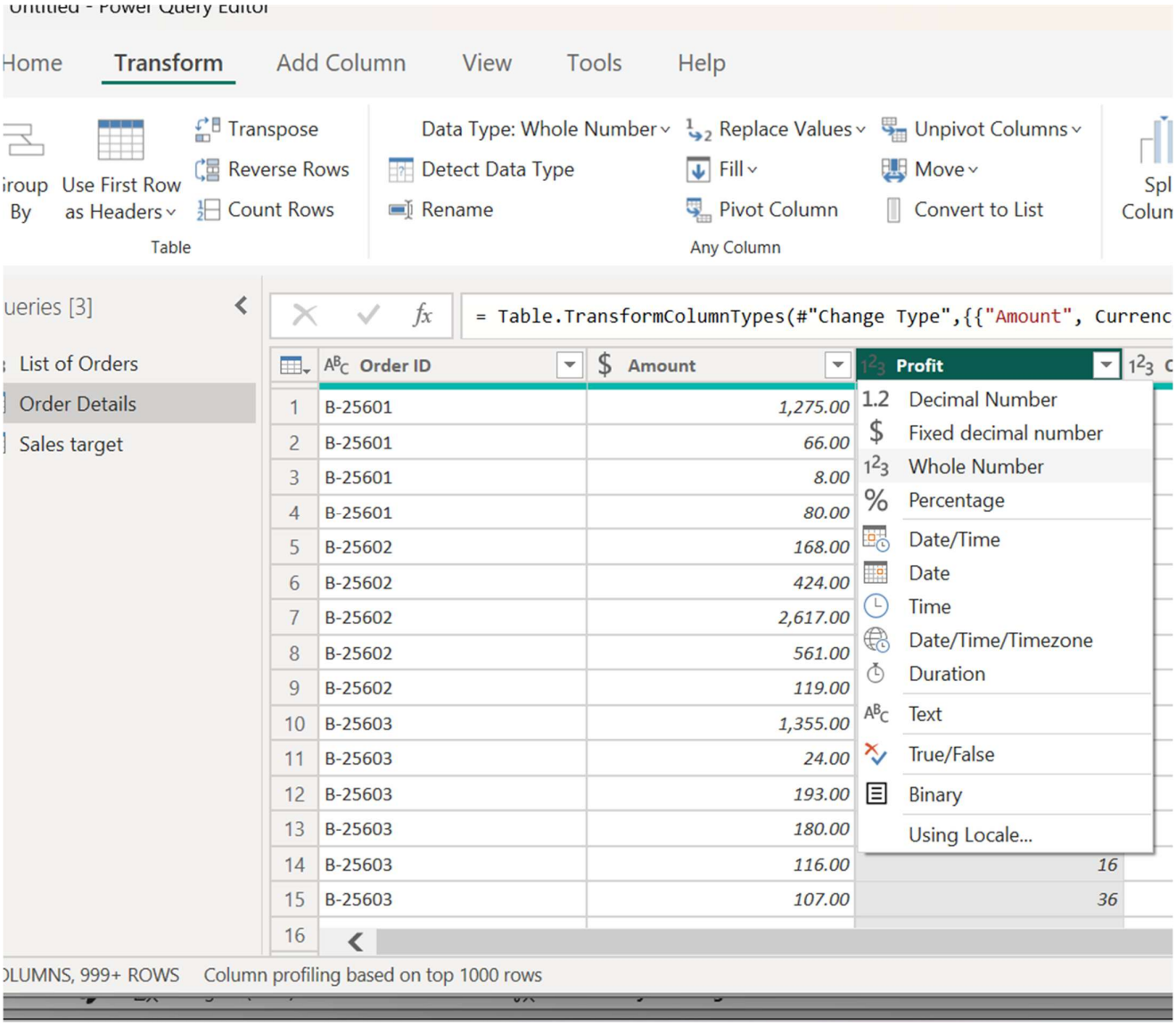
1. Select both the **State** and **City** columns (holding **Ctrl** to select multiple).
2. Go to **Transform** > **Merge Columns**.

3. Set the **Separator** to **Custom**, then enter a comma followed by a space (,).
4. In the **New column name** field, enter **Location** and click **OK**.



To fix the Profit column data type:

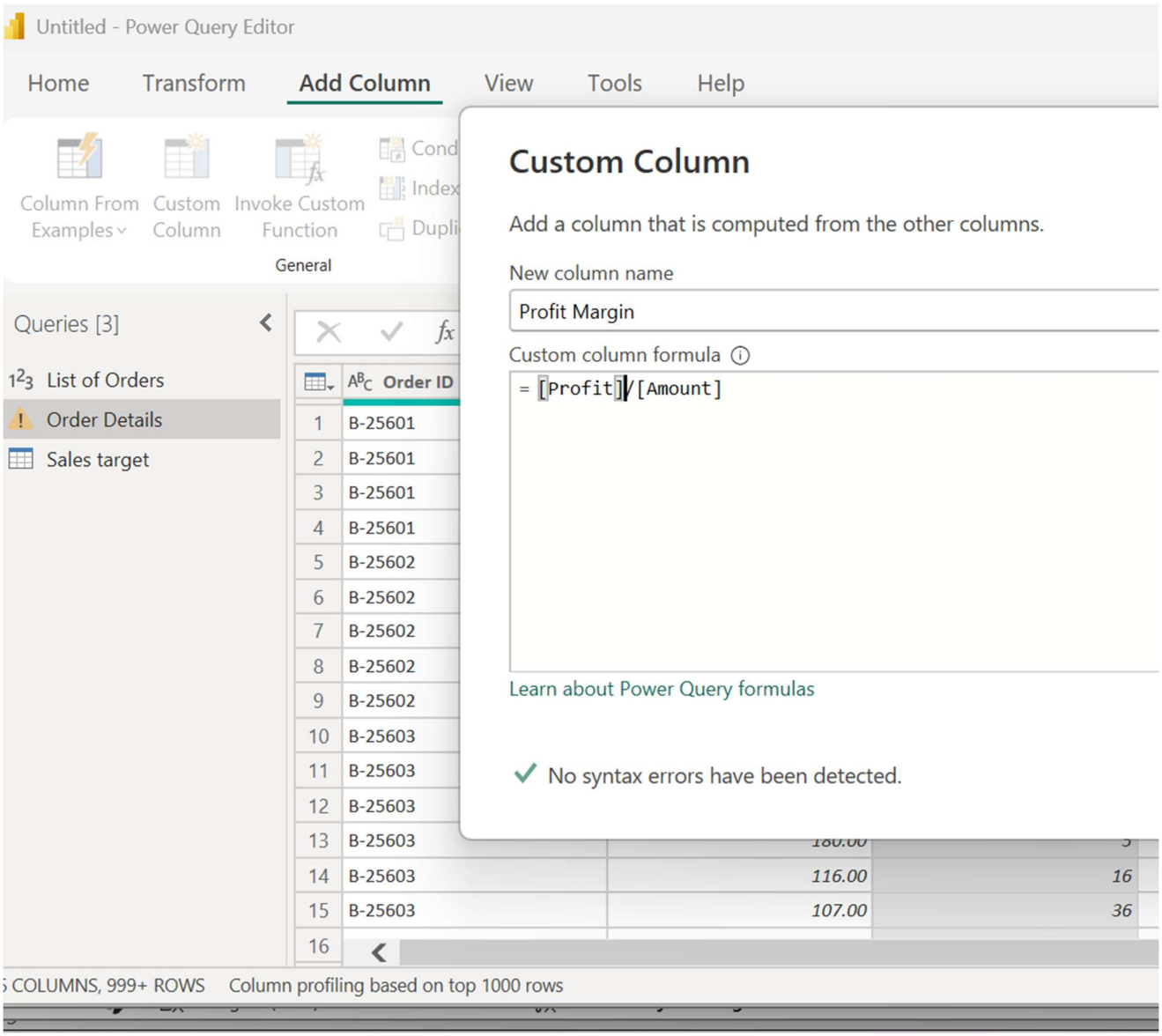
1. Select the **Order Details** query and locate the **Profit** column.
2. Click the **Data Type** icon in the **Profit** column header (currently set to **Text**).
3. Change the data type to **Whole Number** to enable it for future calculations, such as Profit Margin.



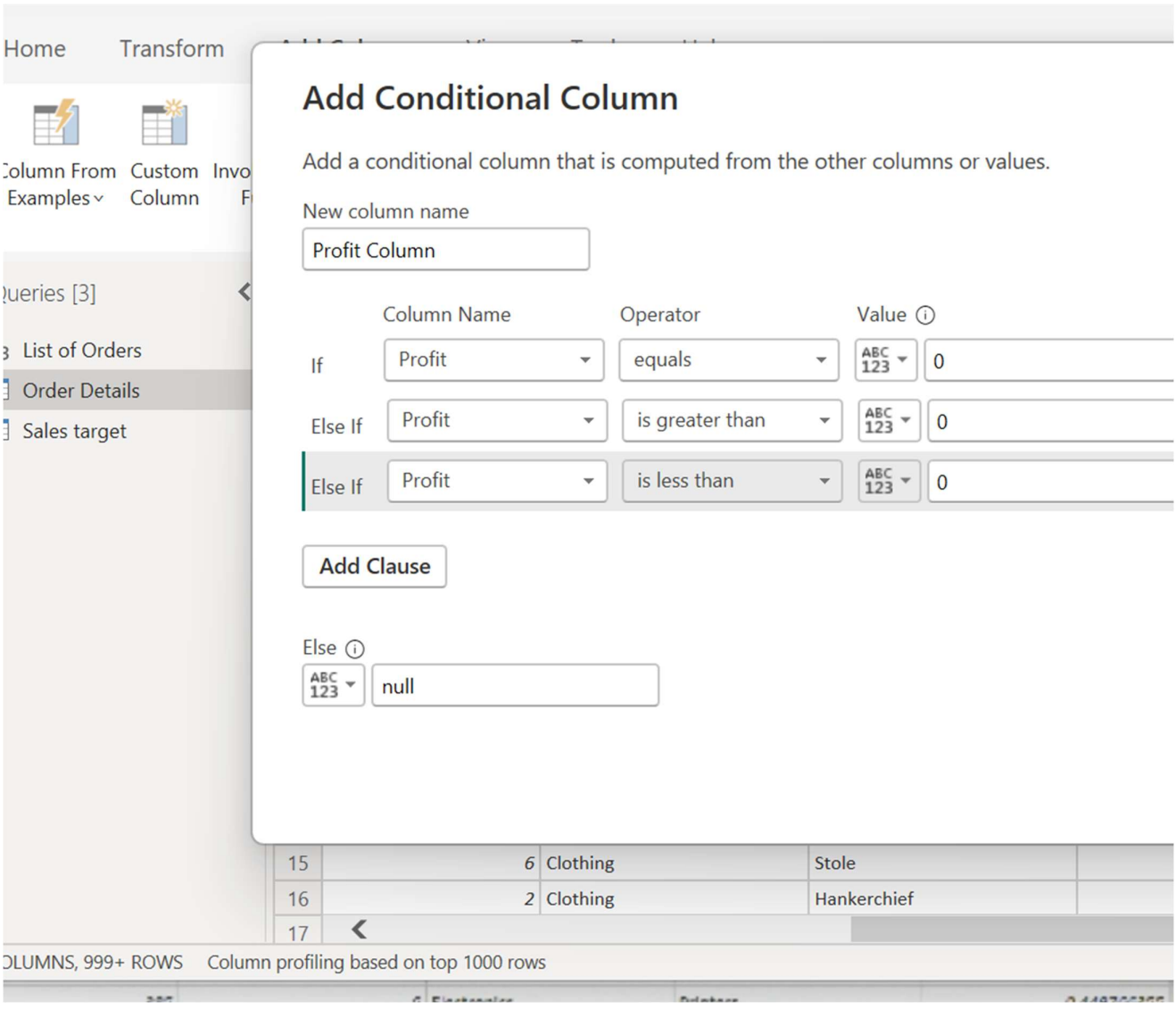
To create the Profit Margin column:

1. Go to the **Add Column** tab and select **Custom Column**.
2. Set the **New column name** to **Profit Margin**.

3. In the formula box, enter [Profit]/[Amount].
4. Click **OK** to create the column and calculate the profit margin values.



Create a conditional field named **Profit Status** by selecting **Add Column > Conditional Column**. Set logic to return **"Break-Even"** if Profit equals 0, **"Profit"** if Profit is greater than 0, and **"Loss"** if Profit is less than 0, then click **OK**.



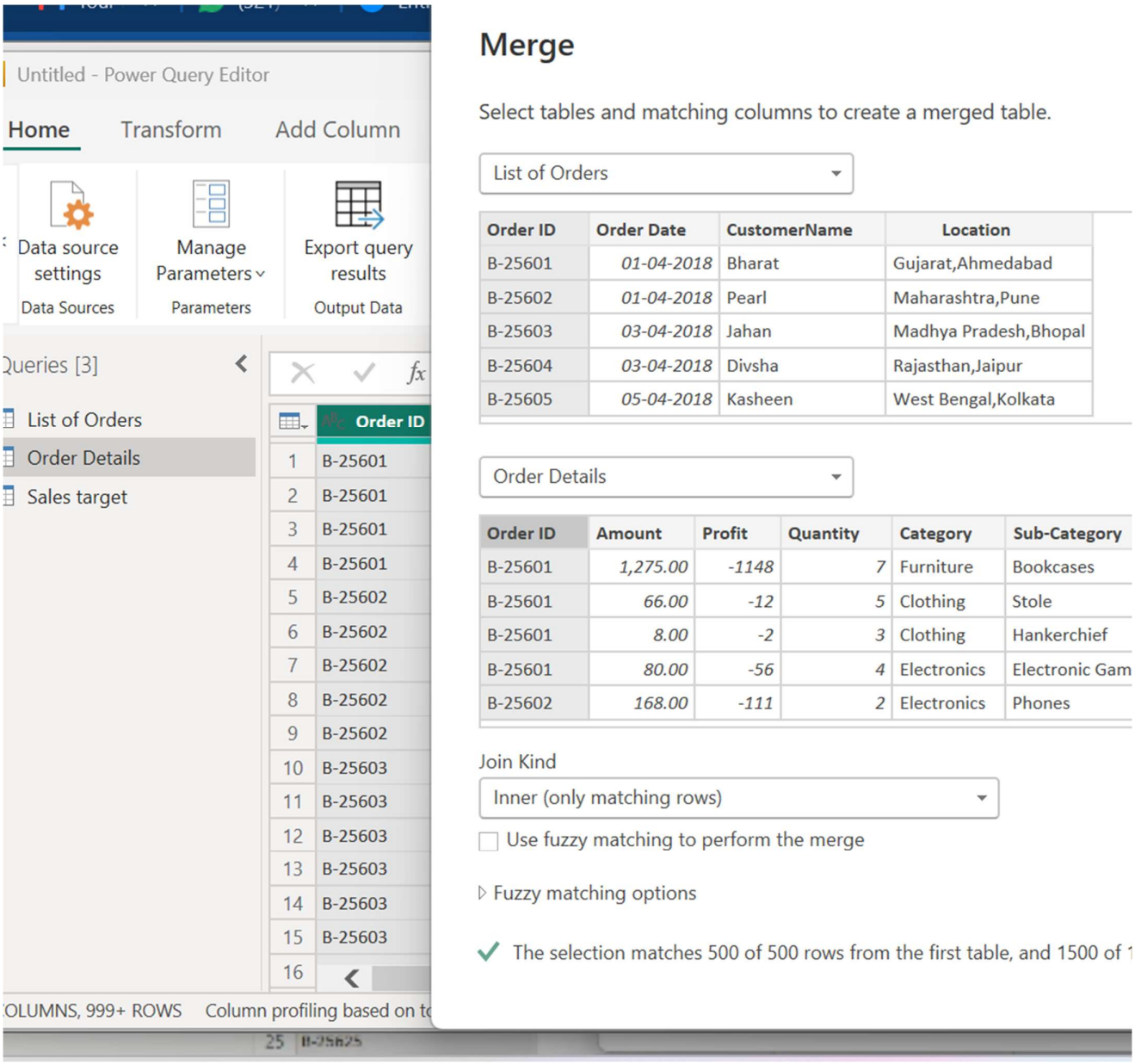
3.Merging Data

To merge the tables into a new query:

1. Go to the **Home** tab and click the drop-down under **Merge Queries**.
2. Select **Merge Queries as New**.
3. This creates a combined table as a separate query while leaving your original two tables unchanged.

To configure the Inner Join:

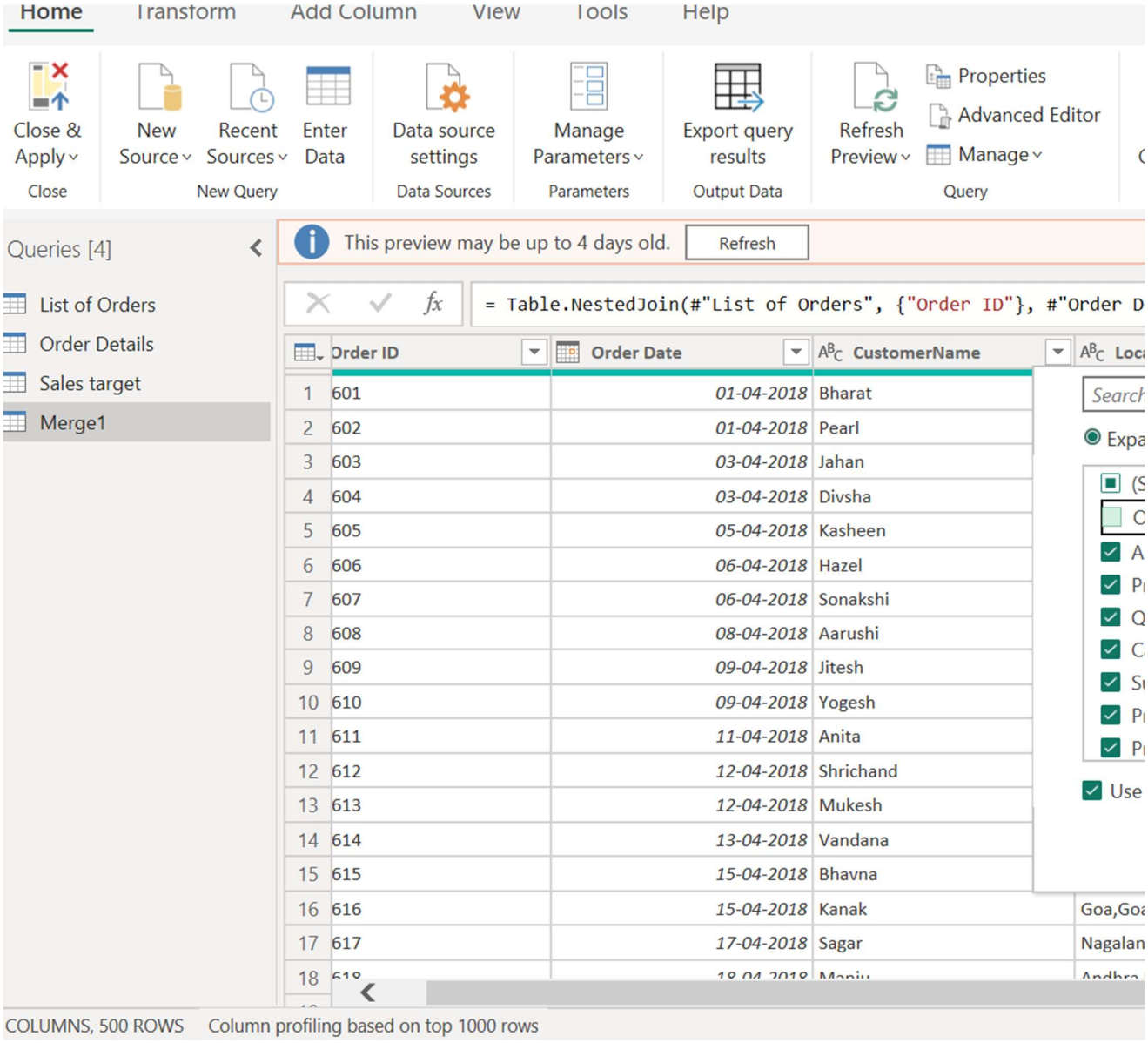
1. Select **List of Orders** as the primary (top) table.
2. Select **Order Details** as the secondary (bottom) table.
3. Click the **Order ID** column in both table previews to set it as the matching key.
4. Set **Join Kind** to **Inner (only matching rows)**.
5. Verify the selection: the confirmation message at the bottom should indicate that all **500 rows** from the first table matched.



Expand the Order Details column

1. Click the **Expand icon** (two opposite-facing arrows) in the **Order Details** column header.

2. Check the boxes for all columns **except Order ID** (to avoid duplicate columns).
3. Click **OK** to expand the nested table.
4. In the **Queries** pane on the left, right-click the query and rename it **Orders Data**.

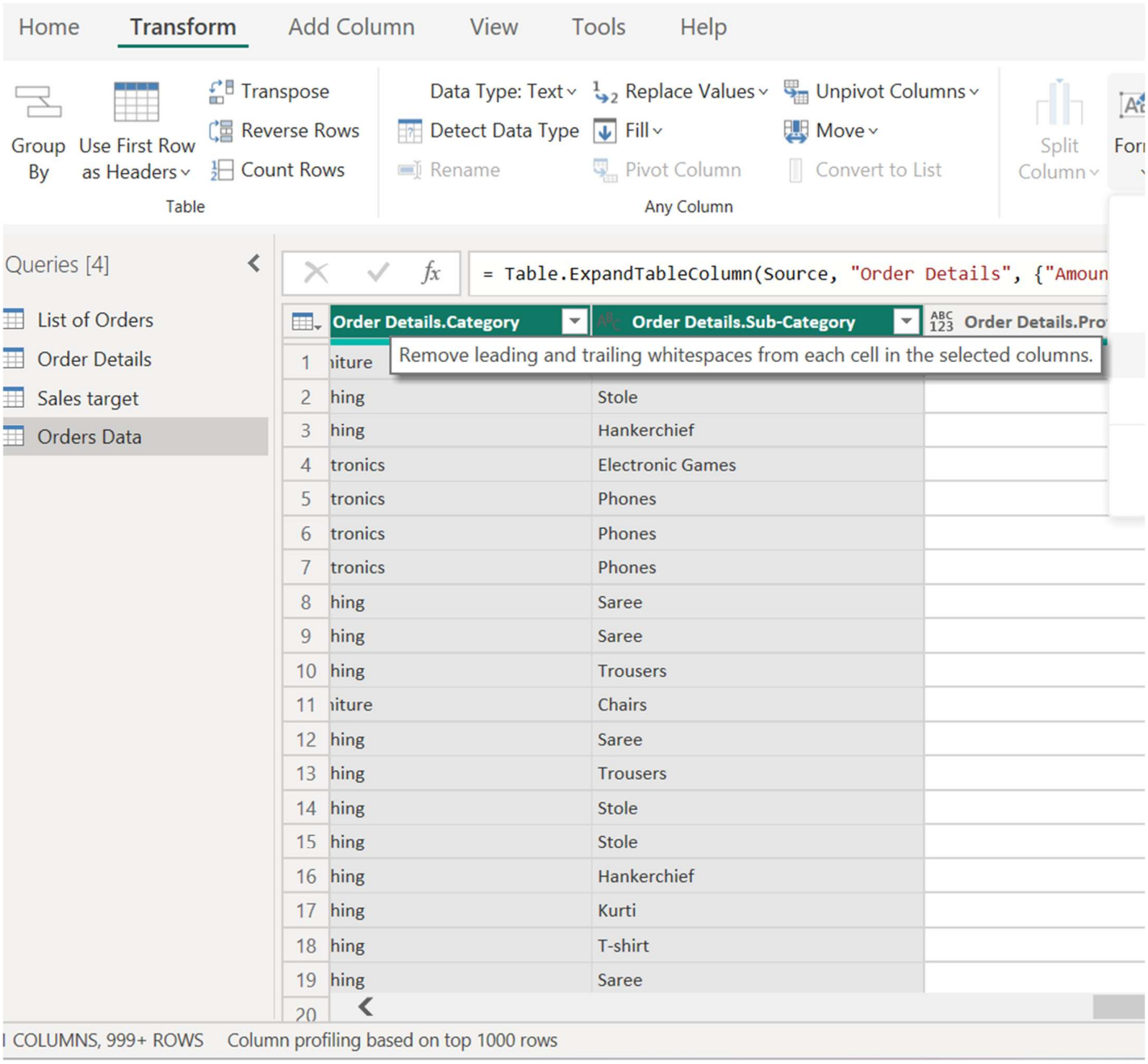


4.Handling Missing and Duplicate Data

To trim extra spaces:

1. In the **Orders Data** query, select all text columns (such as *CustomerName*, *State*, or *City*).
2. Go to the **Transform** tab and select **Format** > **Trim**.

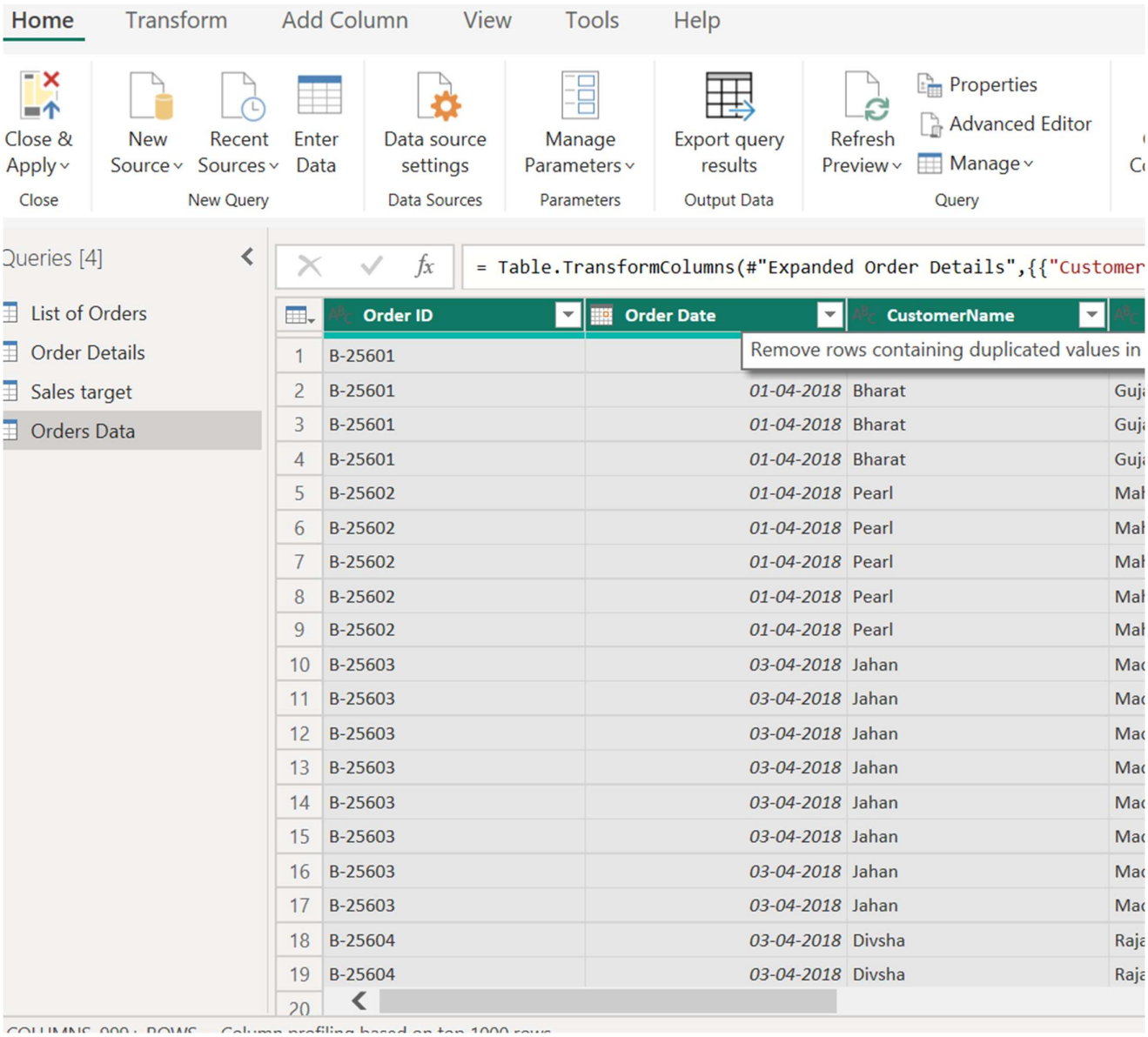
This removes leading and trailing spaces, preventing duplicate records caused by extra spacing.



To remove duplicate rows:

1. Ensure you have completed the **Trimmed Text** and **Cleaned Text** transformation steps.

2. In the **Orders Data** query, navigate to the **Home** tab.
3. Click **Remove Rows** > **Remove Duplicates** to delete any repeated records from the table.

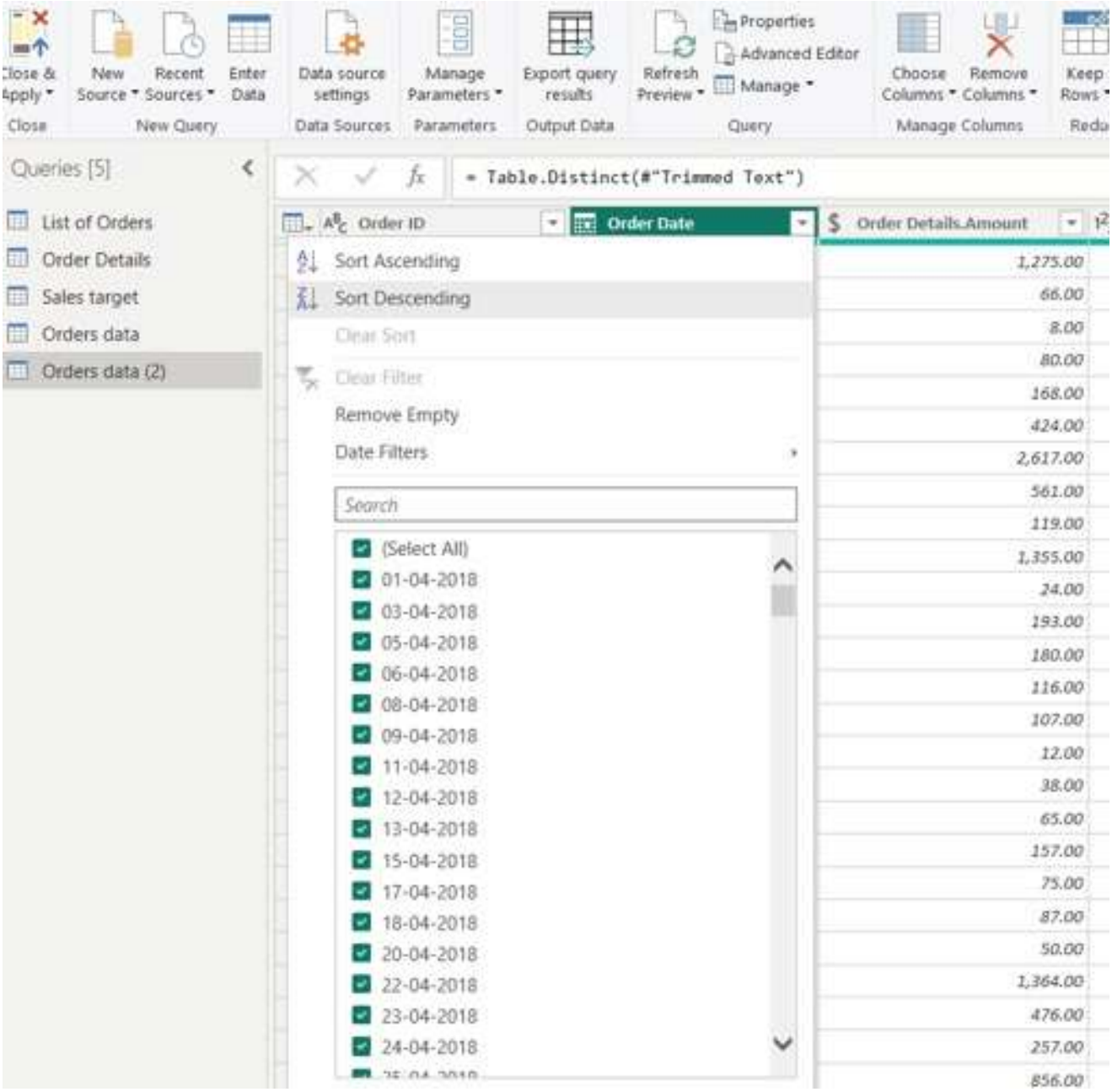


5.Sorting and Filtering Data

To sort Order Date in descending order:

1. **Select the query copy (e.g., Orders Data (2)).**
2. **Click the Filter arrow (≻) on the Order Date column header.**
3. **Select Sort Descending.**

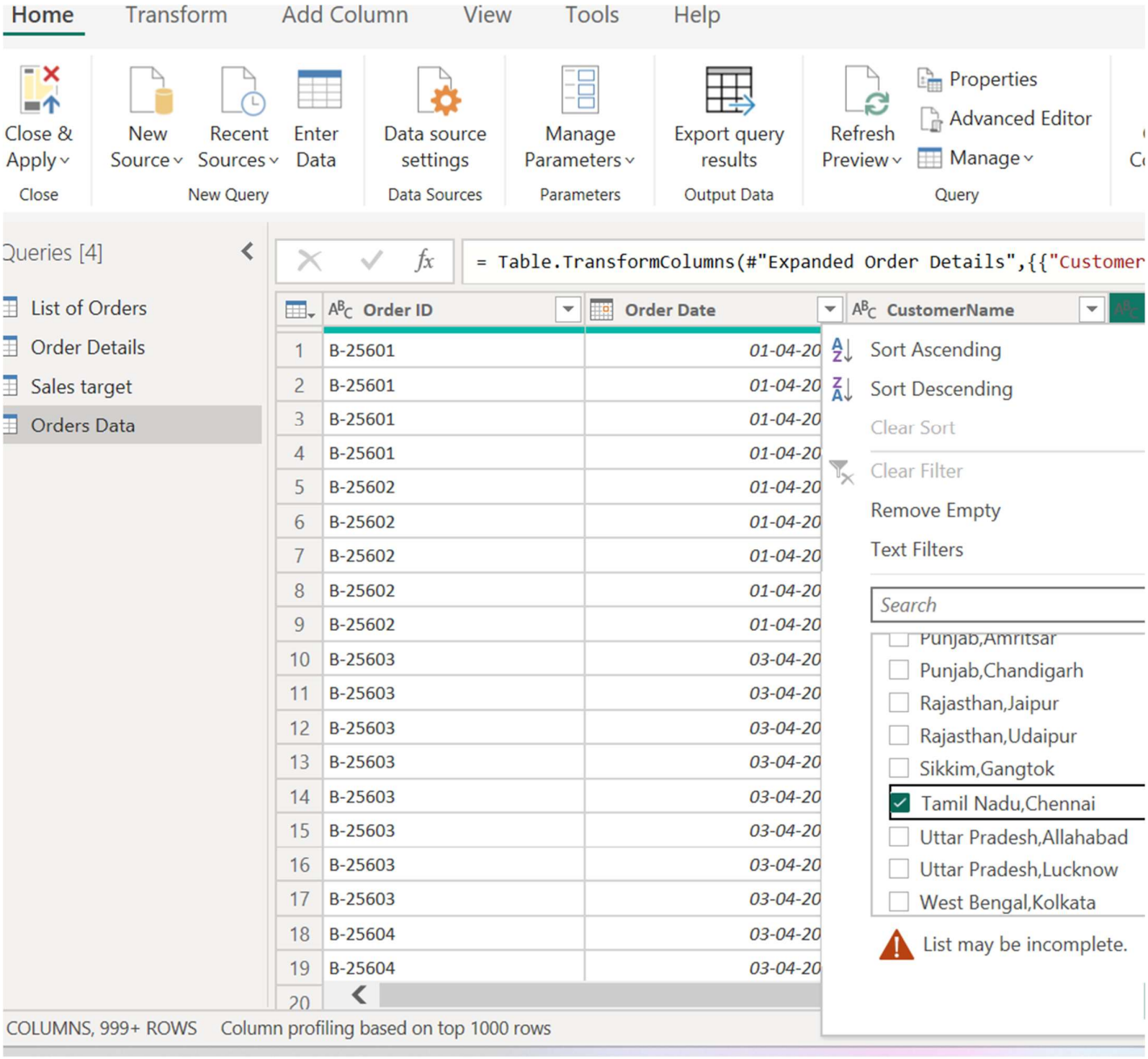
The dataset will reorder so that the most recent orders appear at the top.



To filter the sorted table by location:

1. Verify that the **Sorted Rows** step orders the **Order Date** column in descending order (displaying the latest order, at the top).

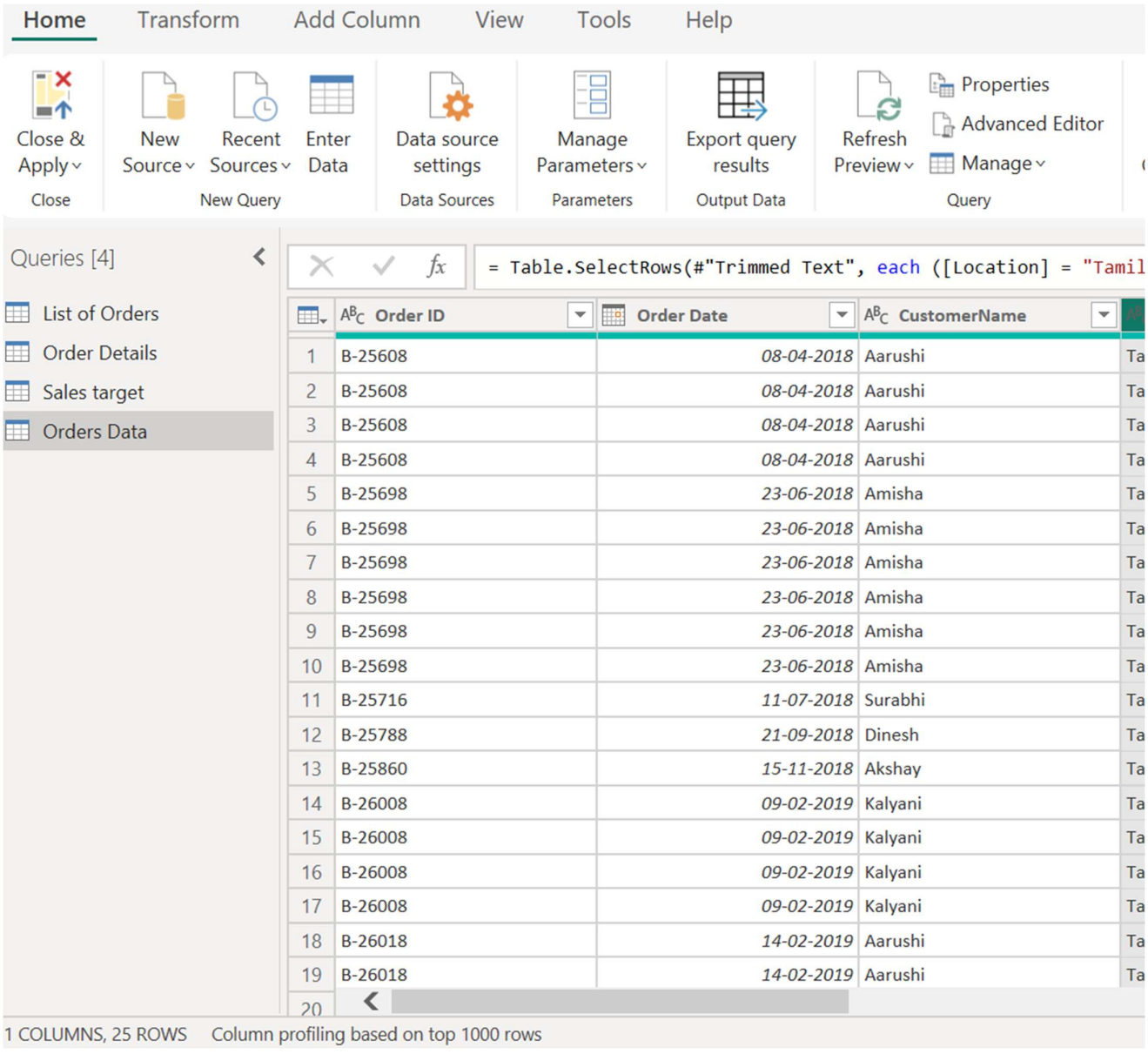
2. Click the **Filter arrow** on the **Location** column header.
3. Uncheck **Select All**, tick only **Tamil Nadu, Chennai**, and click **OK**.



To verify the location filter:

1. Observe the filtered query table to confirm that only rows where **Location** is **Tamil Nadu, Chennai** remain (**54 rows**).

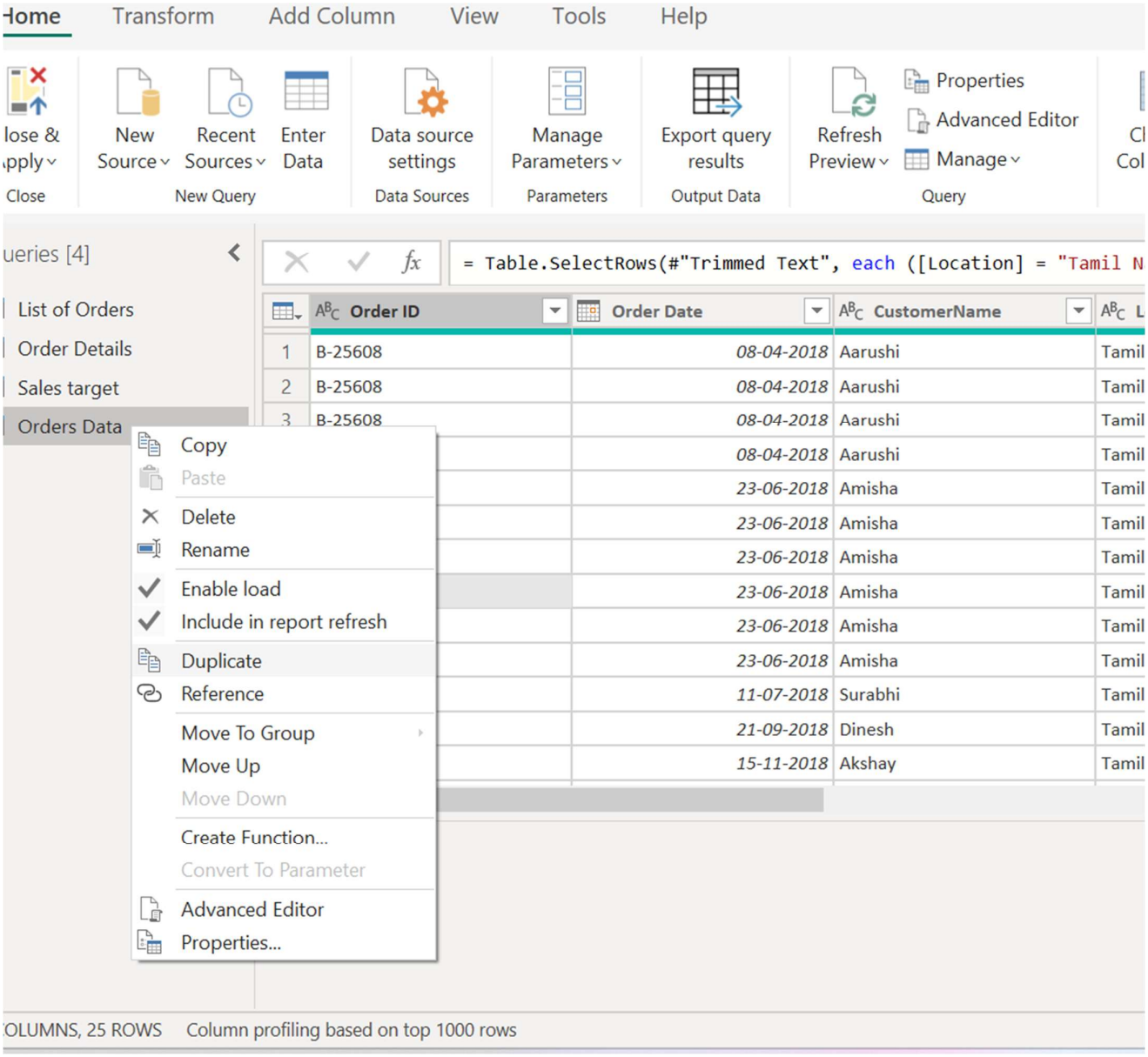
2. Check the **Applied Steps** pane on the right to verify that the **Filtered Rows** step has been added to the sequence.



6.Grouping and Aggregating Data

To duplicate the table for grouping:

1. In the **Queries** pane, right-click the **Orders Data** query.
2. Select **Duplicate**.
3. Perform subsequent grouping operations on this duplicated query to preserve the original dataset unchanged.

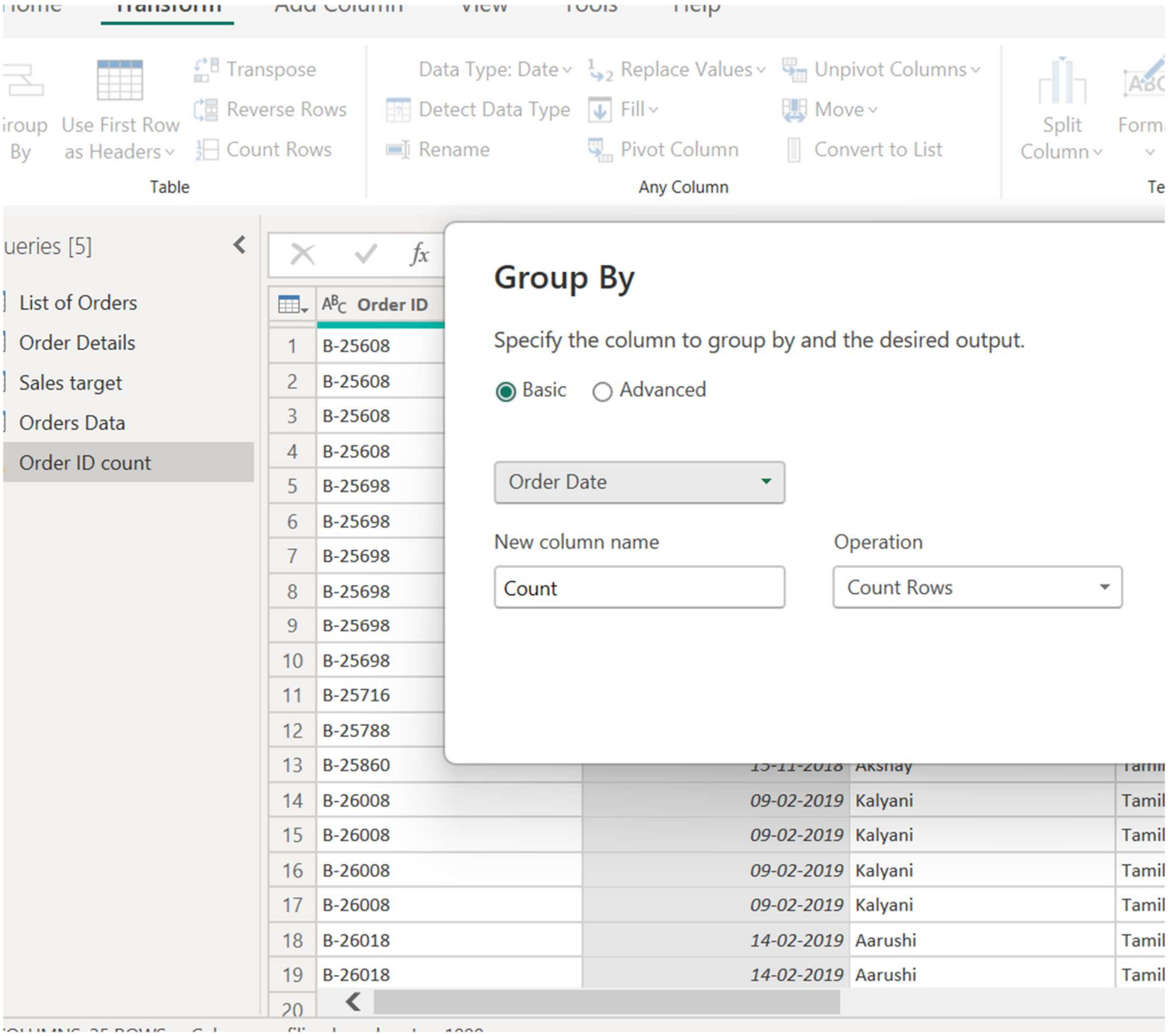


To count the number of items per Order ID:

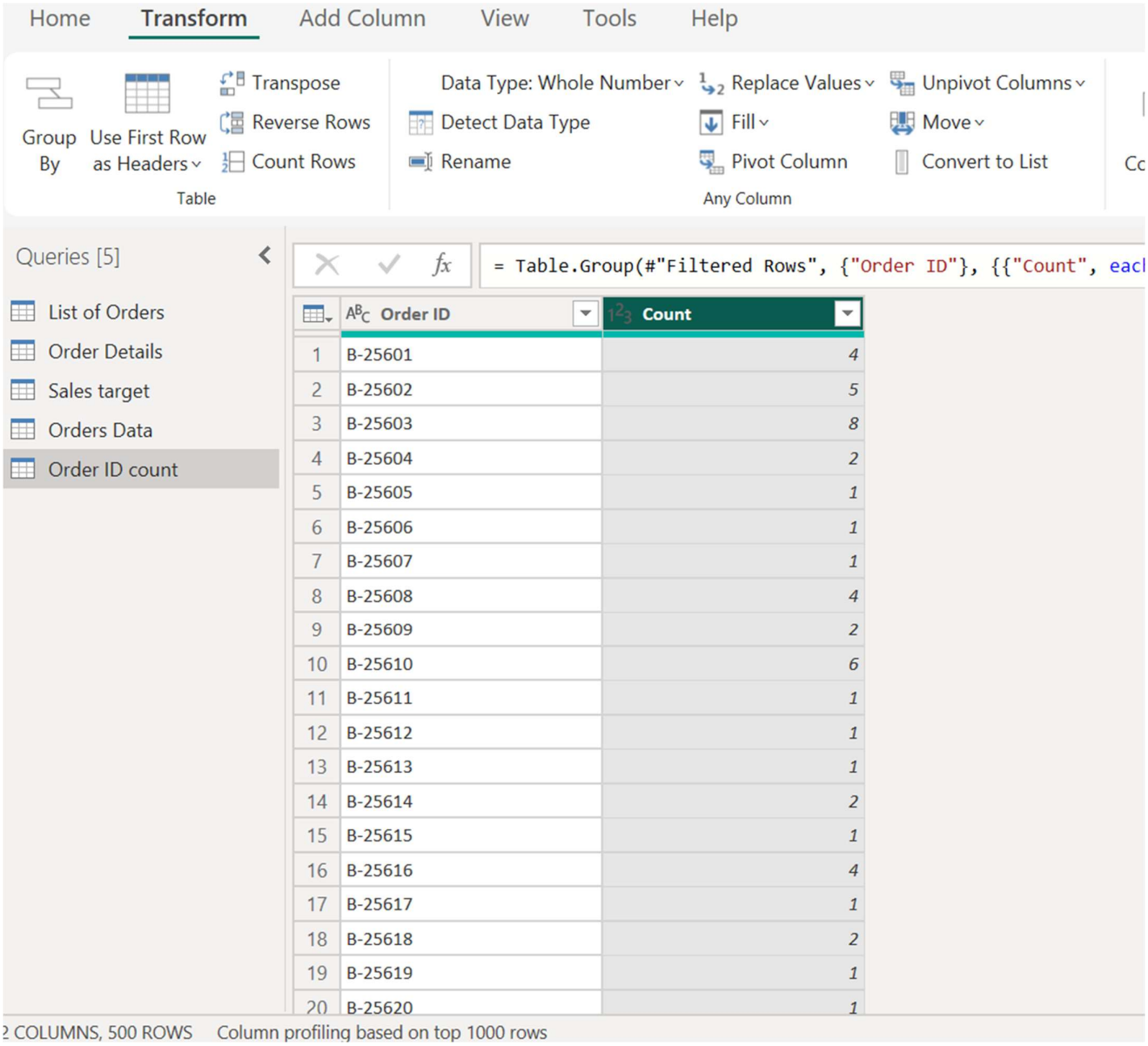
1. In the duplicated query, navigate to the **Transform** tab and select **Group By**.

2. In the **Group By** dialog, set the **Group by** column to **Order ID**.
3. Set the **New column name** to **Count**.
4. Set the **Operation** to **Count Rows** and click **OK**.
5. Rename this query to **Count of Order ID**.

The output will list each Order ID alongside its item count (e.g., Order B-25601 shows 4 items).



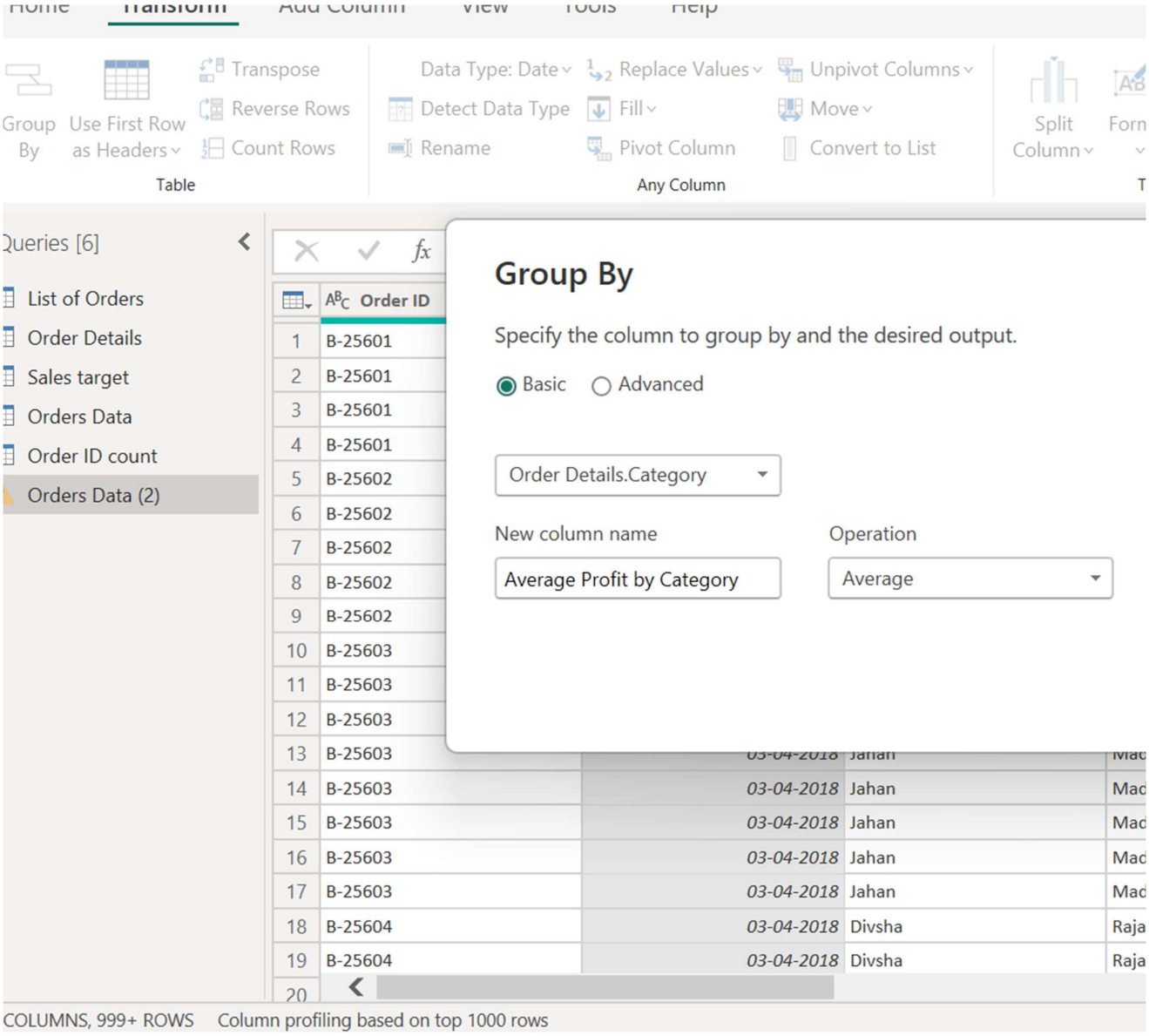
	Order ID			
1	B-25608			
2	B-25608			
3	B-25608			
4	B-25608			
5	B-25698			
6	B-25698			
7	B-25698			
8	B-25698			
9	B-25698			
10	B-25698			
11	B-25716			
12	B-25788			
13	B-25860	15-11-2018	Akshay	Tamil
14	B-26008	09-02-2019	Kalyani	Tamil
15	B-26008	09-02-2019	Kalyani	Tamil
16	B-26008	09-02-2019	Kalyani	Tamil
17	B-26008	09-02-2019	Kalyani	Tamil
18	B-26018	14-02-2019	Aarushi	Tamil
19	B-26018	14-02-2019	Aarushi	Tamil
20				



To calculate average profit by category:

1. Select another copy of the **Orders Data** query.
2. Go to the **Transform** tab and click **Group By**.
3. Set the **Group by** column to **Category**.

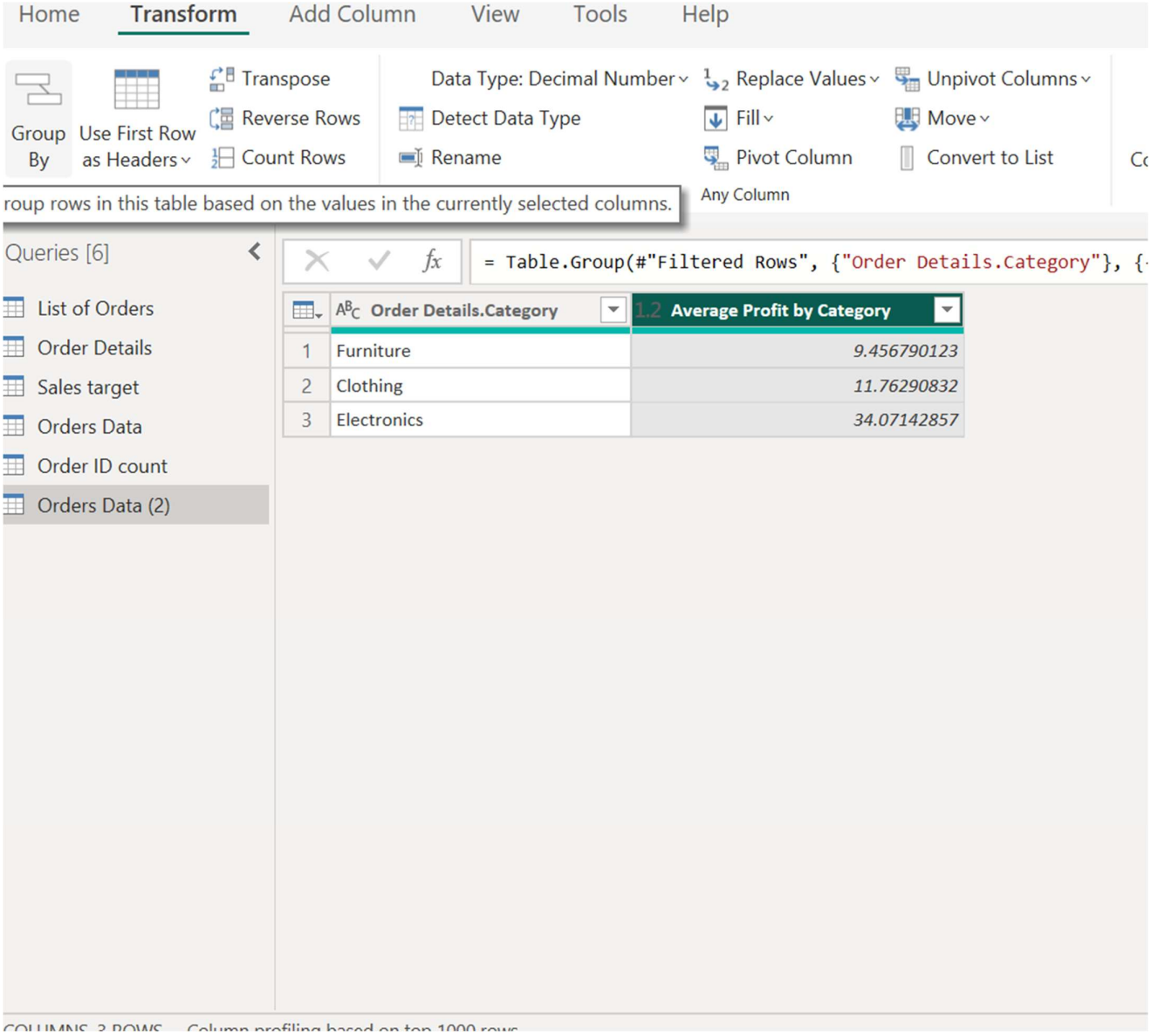
4. Set the **New column name** to **Avg. Profit By Category**.
5. Set the **Operation** to **Average**.
6. Set the **Column** field to **Profit**.
7. Click **OK** to aggregate the dataset.



To verify the average profit results:

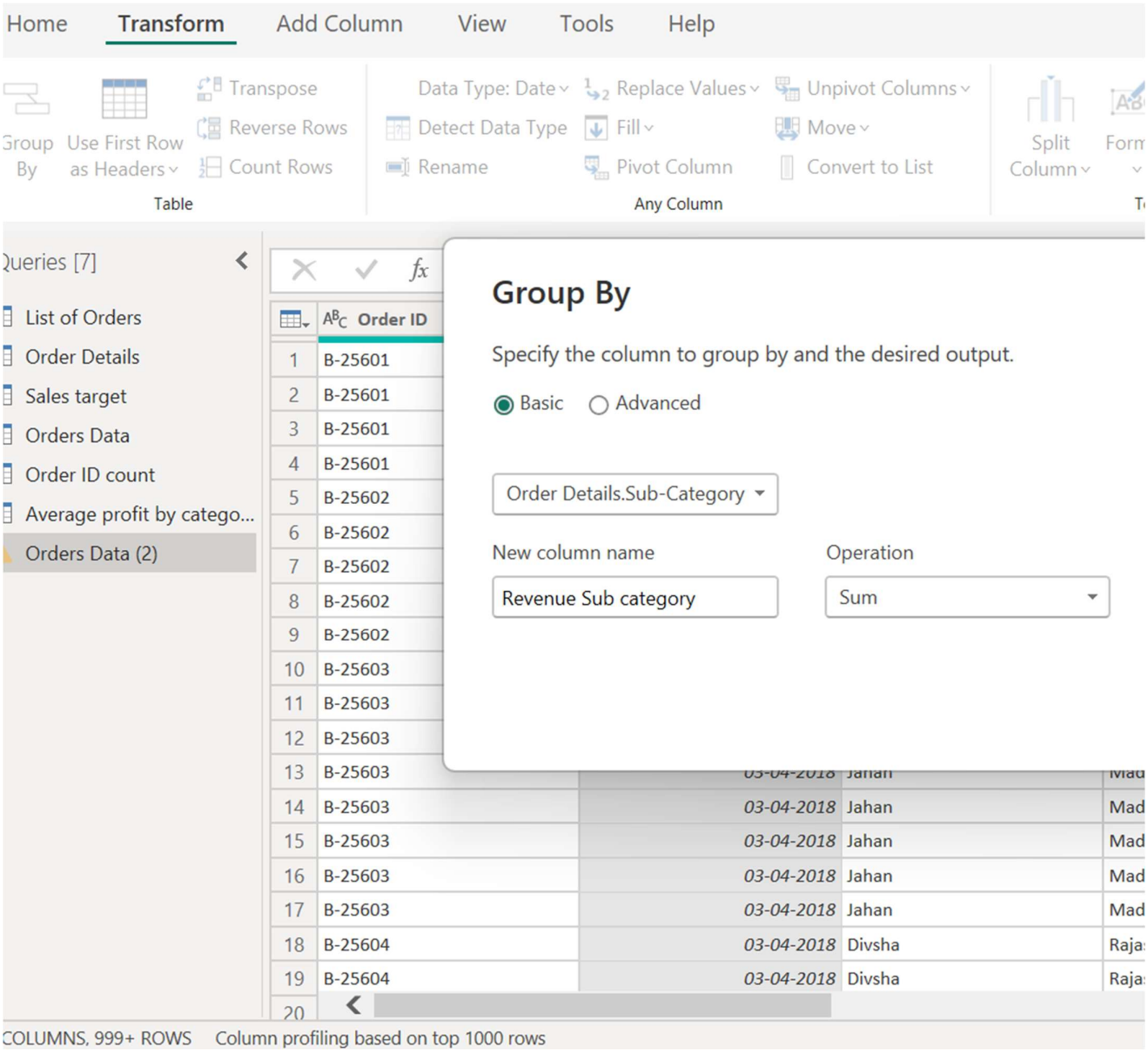
1. Review the aggregated output in the **Avg. Profit By Category** query.
2. Confirm that each category displays its calculated average profit:

- **Furniture:** ~9.46
- **Clothing:** ~11.76
- **Electronics:** ~34.07



To calculate total revenue by Sub-Category:

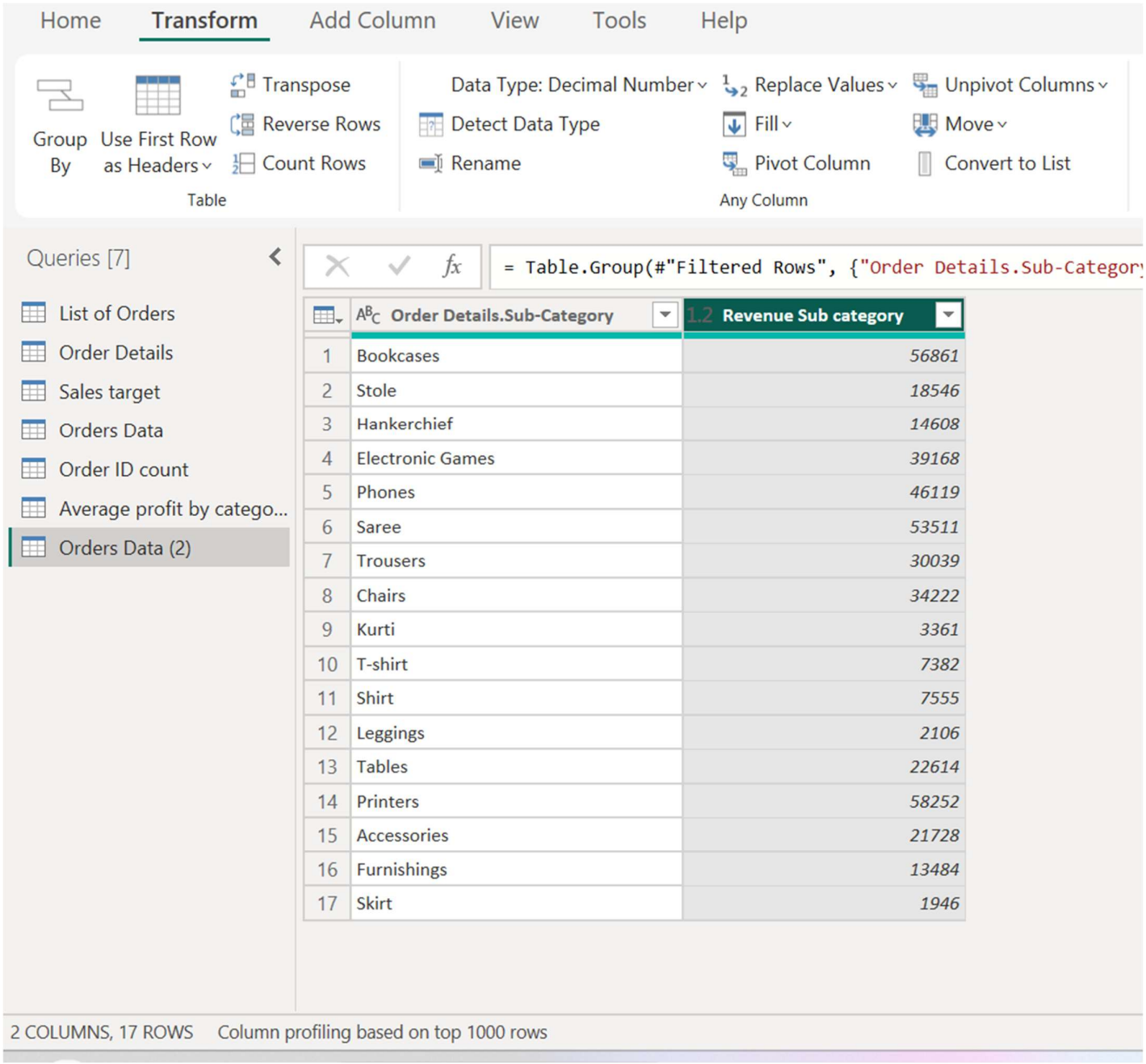
1. Select the third copy of the **Orders Data** query.
2. Navigate to the **Transform** tab and click **Group By**.
3. Set the **Group by** column to **Sub-Category**.
4. Set the **New column name** to **Revenue by S-Cat**.
5. Set the **Operation** to **Sum**.
6. Set the **Column** field to **Amount**.
7. Click **OK** to generate the aggregated revenue summary.



To verify the sub-category revenue results:

1. Review the output in the **Revenue by S-Cat** query.

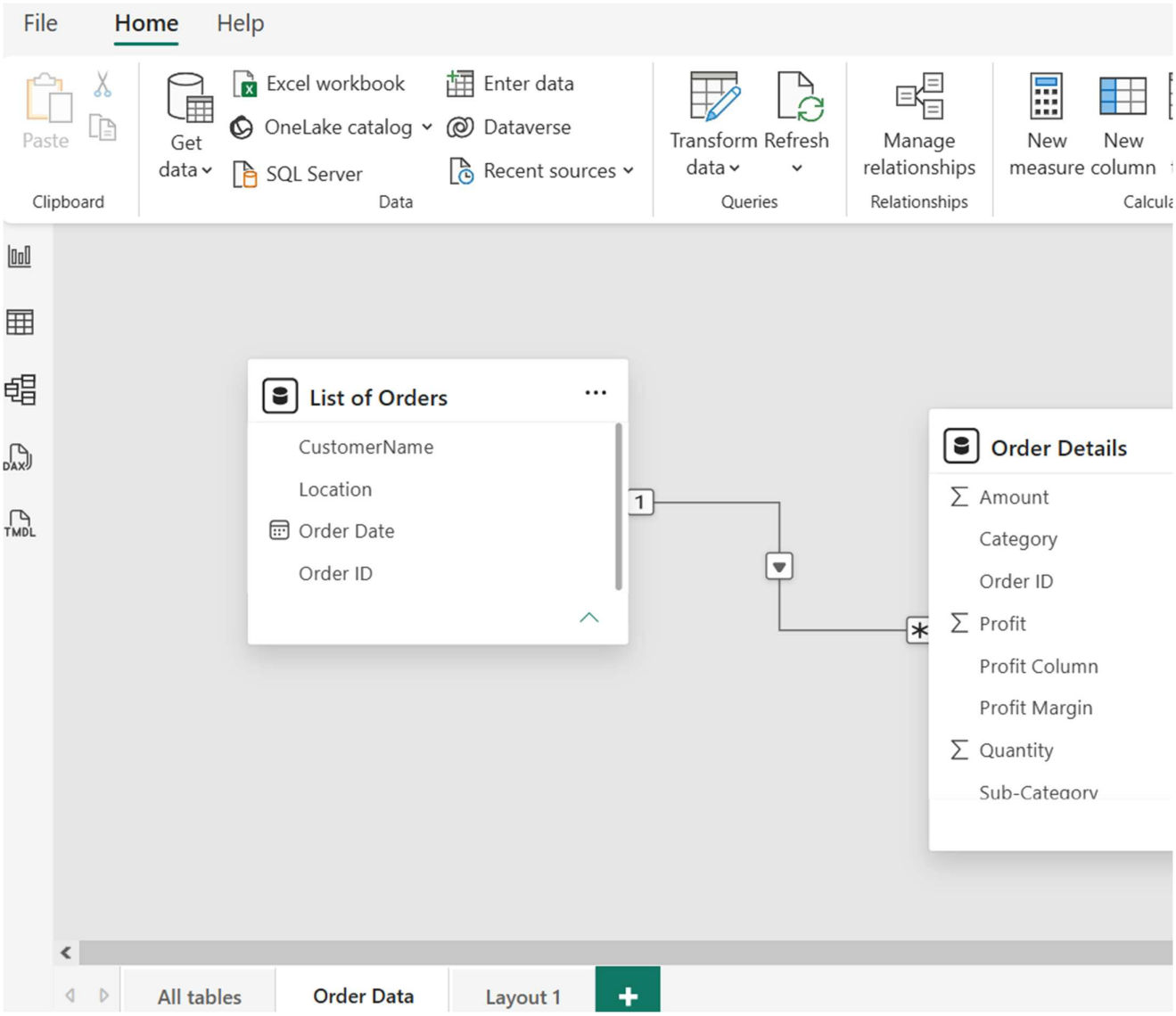
2. Confirm that total sales figures are generated for all **17 sub-categories**.
3. Verify top sub-category totals, such as:
    - **Printers:** 58,252
    - **Bookcases:** 56,861
    - **Saree:** 53,511



7.Data Modeling

To verify the relationship between List of Orders and Order Details:

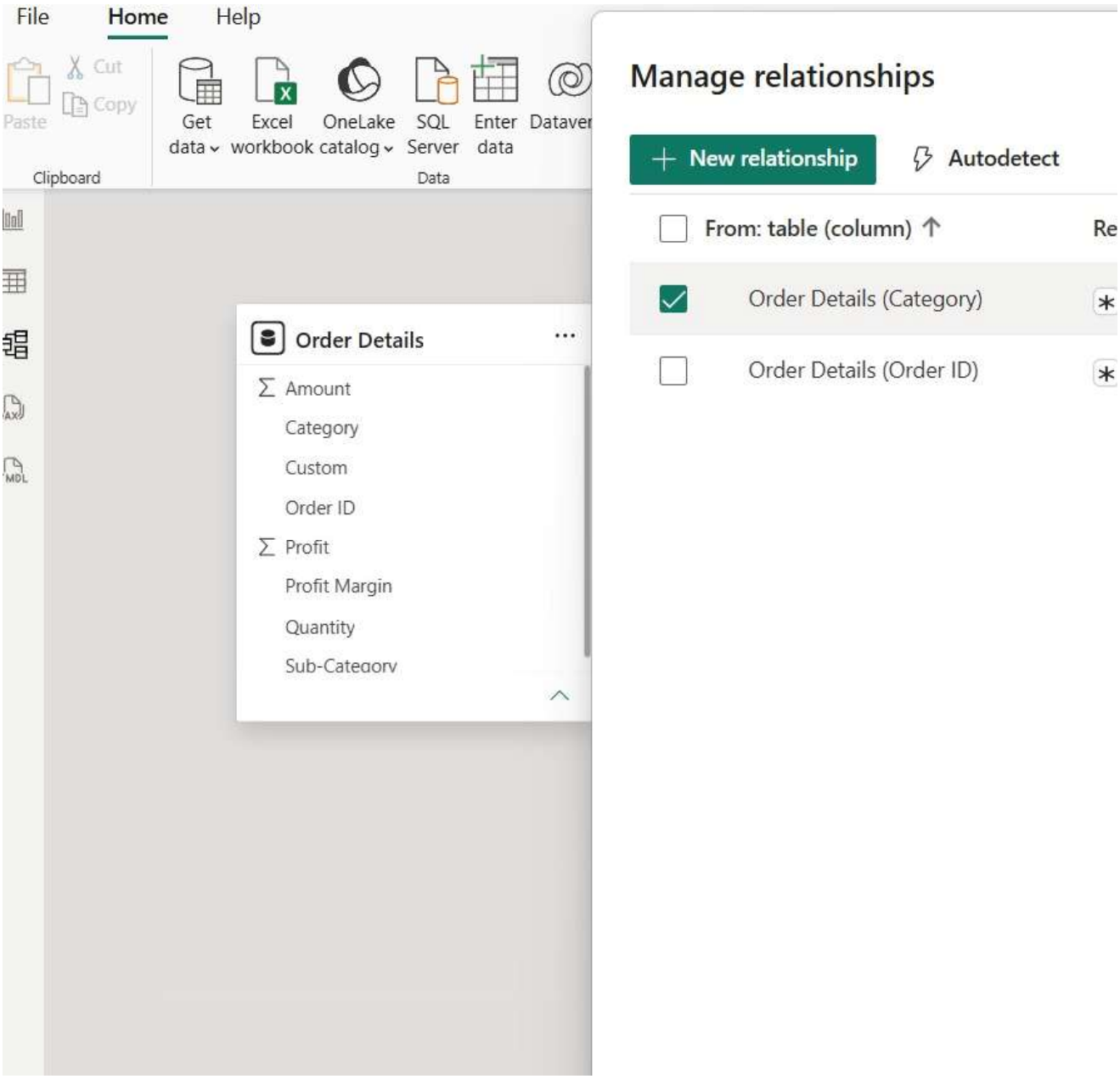
1. **In Power Query, click Close & Apply to load changes into the model.**
2. **Switch to the Model view using the left navigation pane.**
3. **Locate the link connecting List of Orders and Order Details.**
4. **Confirm that the relationship is established using the Order ID field.**
5. **Verify the 1-to-Many (1:*) relationship cardinality:**
    - **1 on the List of Orders side indicates each Order ID is unique.**
    - ******* on the Order Details side indicates an Order ID can appear multiple times across individual line items.**



To review and manage model relationships:

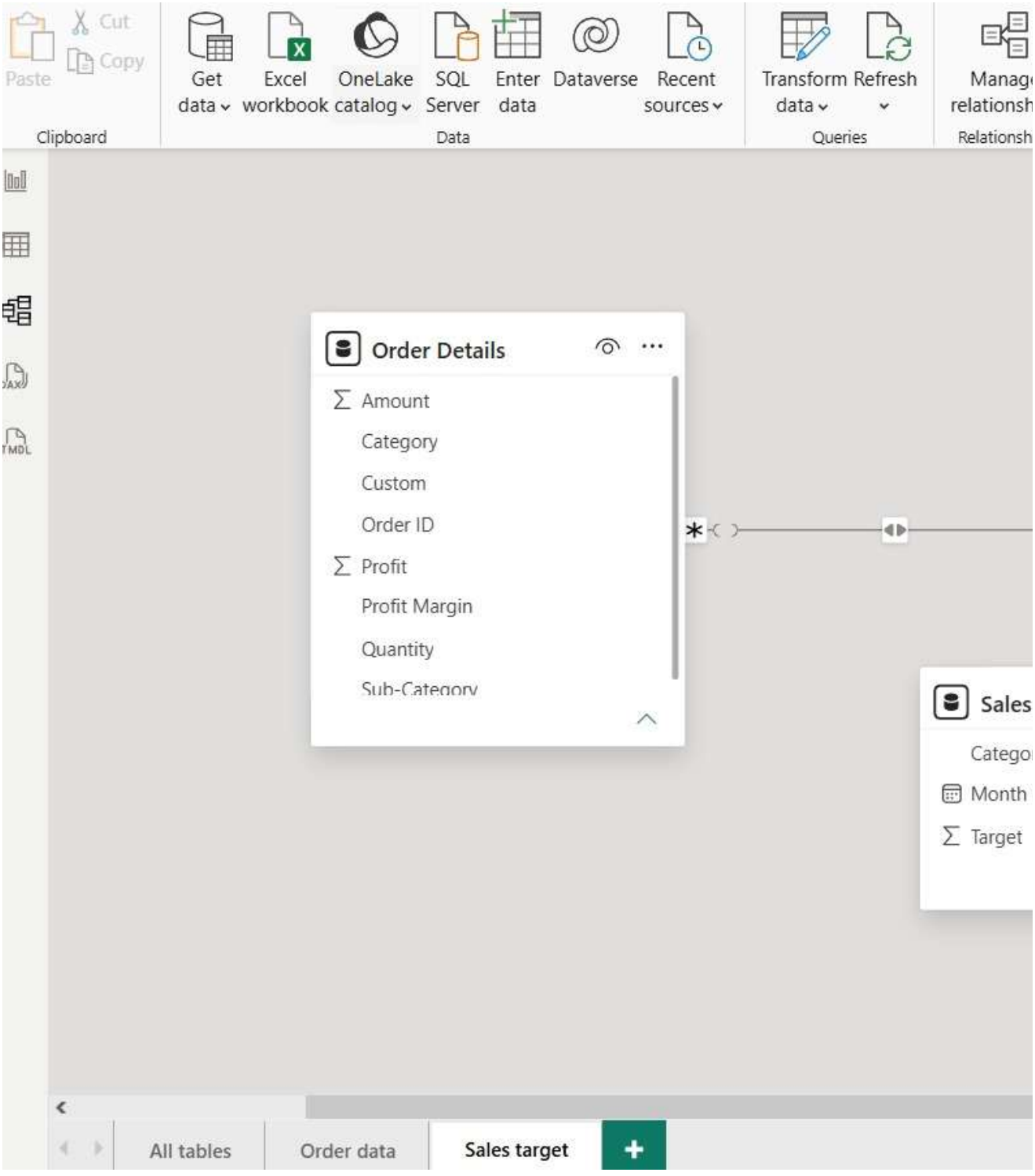
1. **Go to the Home tab and select Manage relationships.**
2. **Locate and select the relationship connecting Order Details and Sales target.**

3. **Verify that the link joins the Category field from both tables and that the Active checkbox is checked.**



To review the Order Details and Sales Target relationship:

1. **In the Model view, locate the link between Order Details and Sales target.**
2. **Confirm that both tables are connected using the Category column.**
3. **Verify the Many-to-Many (M:N or *:*) cardinality:**
    - **Category contains duplicate values in Order Details (multiple sales records per category).**
    - **Category contains duplicate values in Sales target (multiple target records per category, e.g., monthly target entries).**


Cut
Copy
Paste
Clipboard
Get data
Excel workbook
OneLake catalog
SQL Server
Enter data
Dataverse
Recent sources
Data
Transform data
Refresh
Queries
Manage relationsh
Relationsh
Order Details
Amount
Category
Custom
Order ID
Profit
Profit Margin
Quantity
Sub-Category
Sales
Month
Target
All tables
Order data
Sales target

Summary:

This assignment demonstrates an end-to-end Power BI data workflow, covering data importing, cleansing and transformation, table merging, and data model design using the Orders dataset.